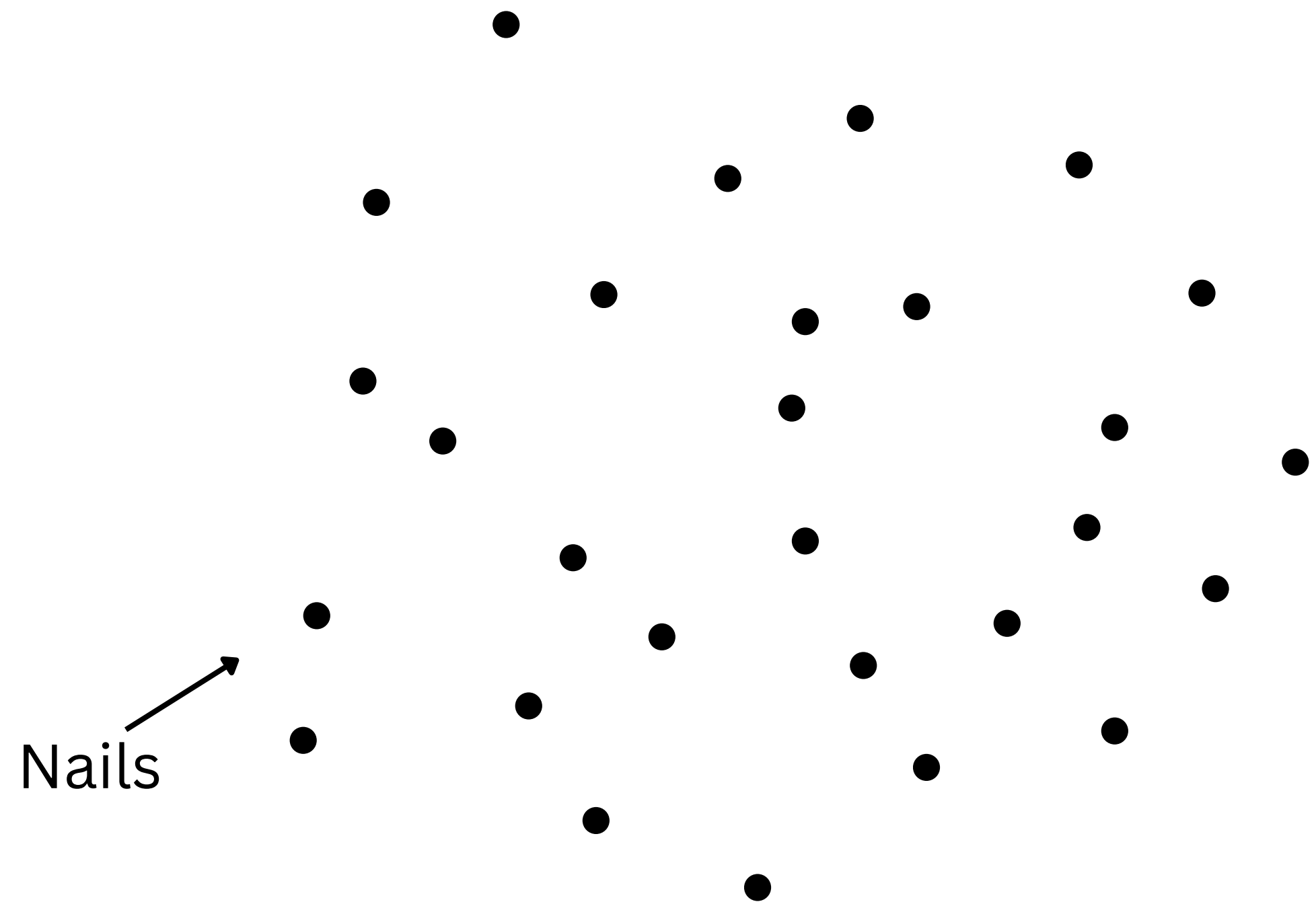
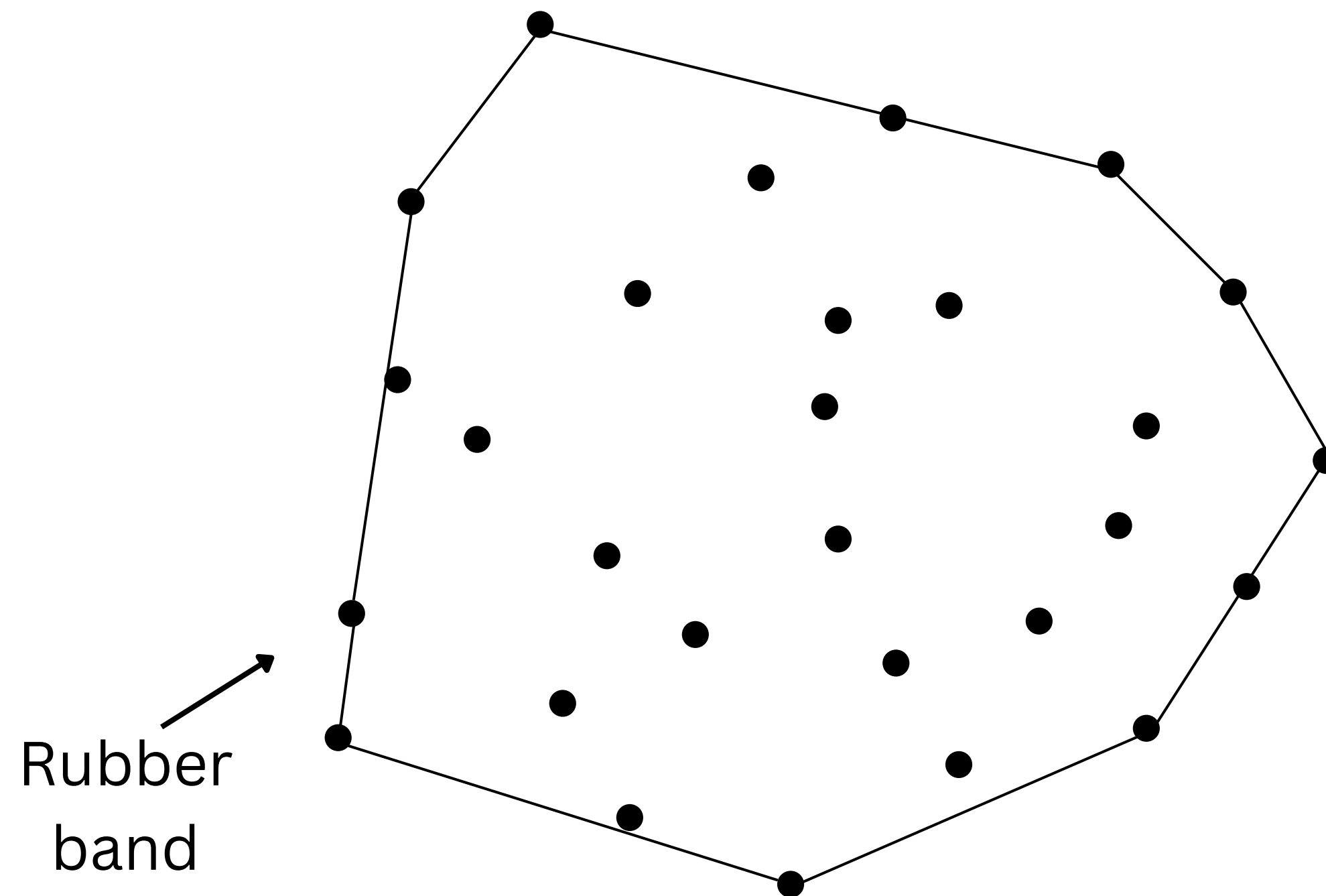


Andrew's Monotone Chain Algorithm

What is a Convex Hull?

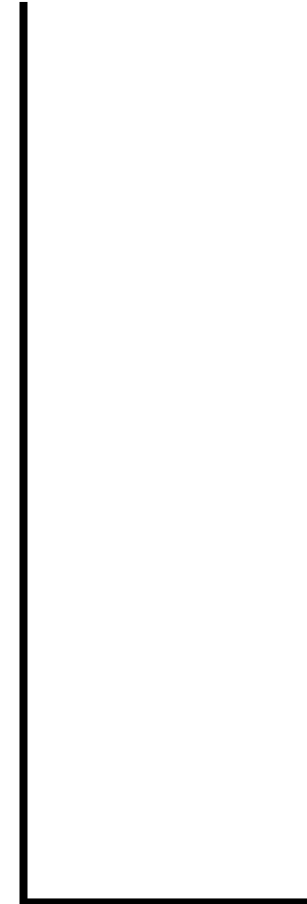
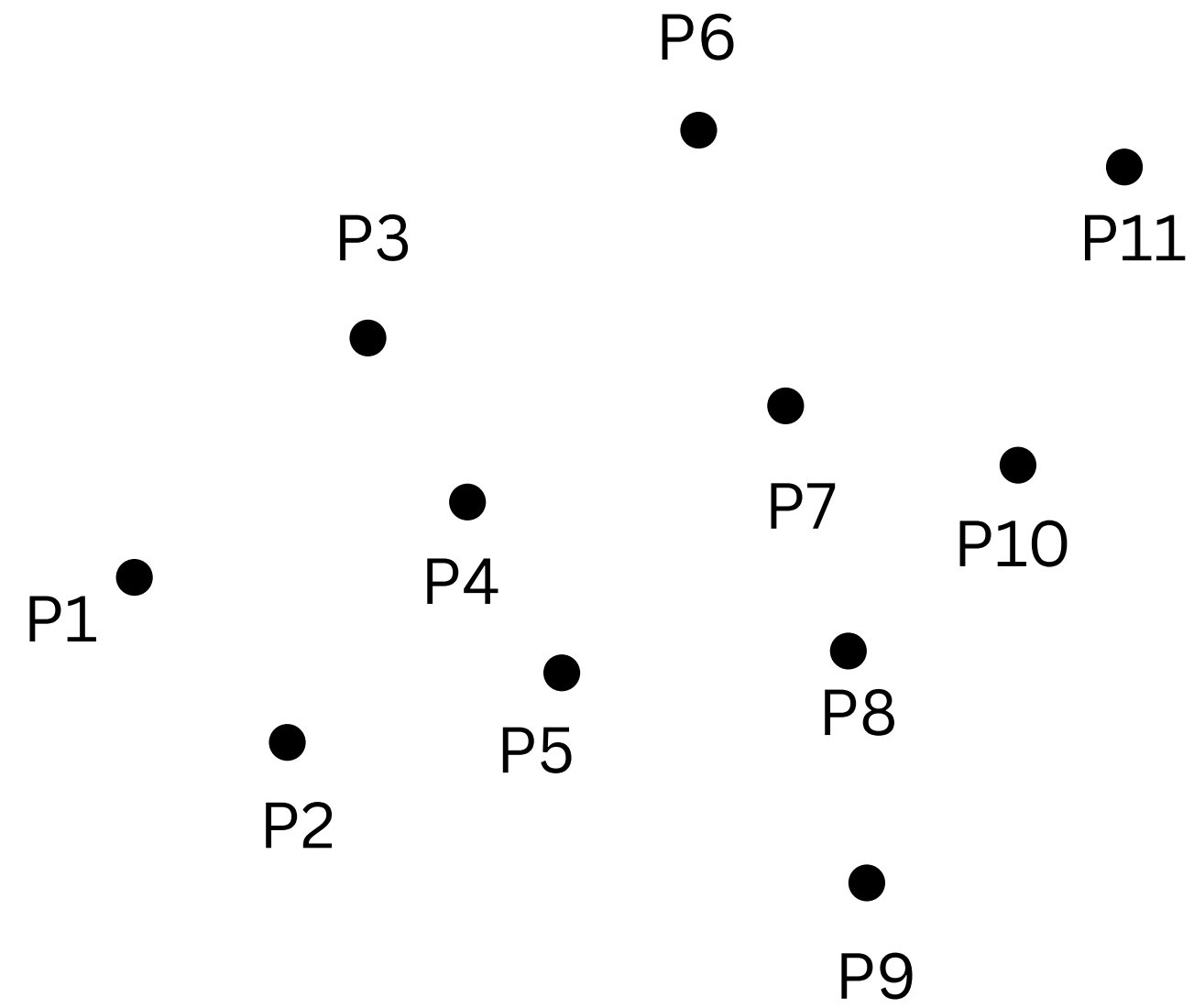




The Data Structures

Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 P11

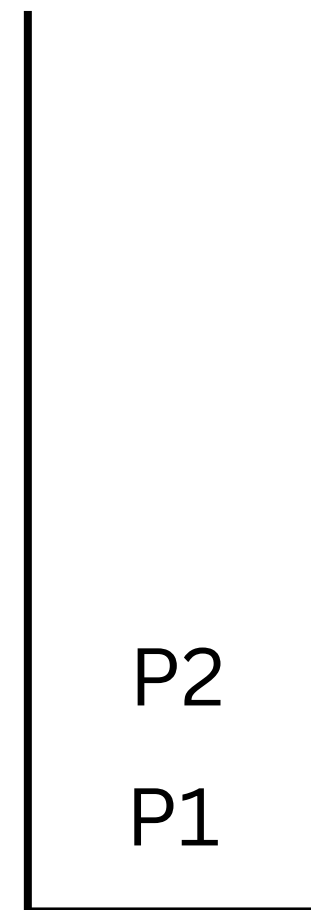
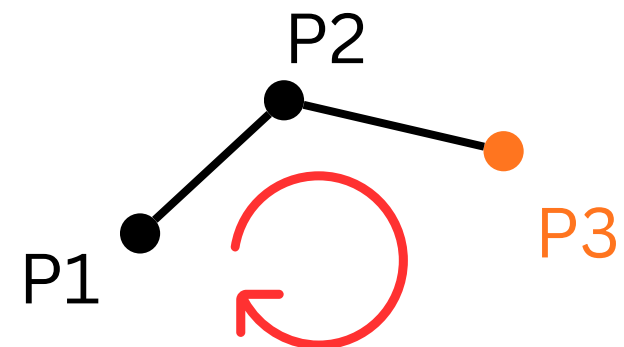
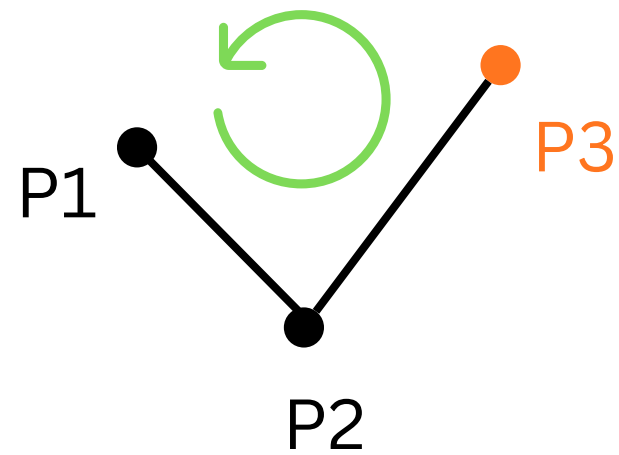


Stack

The Algorithm

Array

P1 P2 **P3** P4 P5 P6 P7 P8 P9 P10 P11



Stack

If the top 2 points in the stack with the point under consideration when taken in that order (P1, P2 and P3 here) make a counter-clockwise turn, push the point onto the stack, else pop the stack and keep repeating this procedure while the last 3 points do not make a counter-clockwise turn.

Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 P11



P1



P2



P3



P4



P5



P6



P7



P8



P9



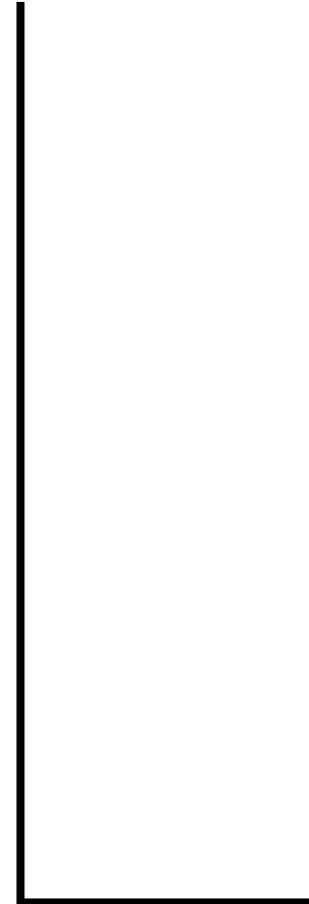
P10



P11

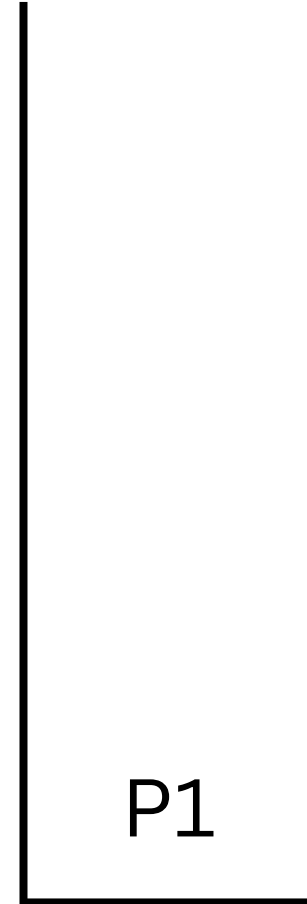
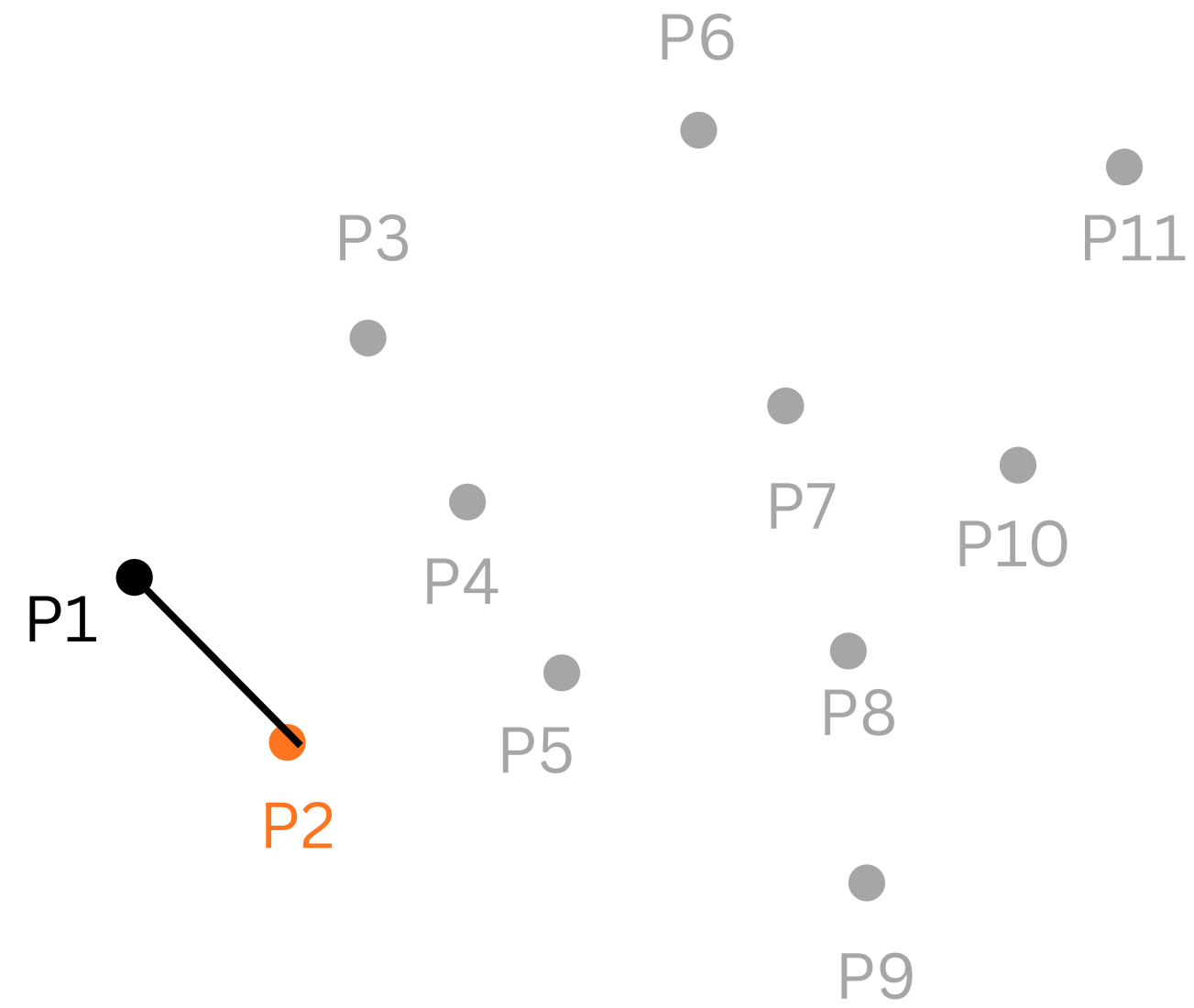


Stack



Array

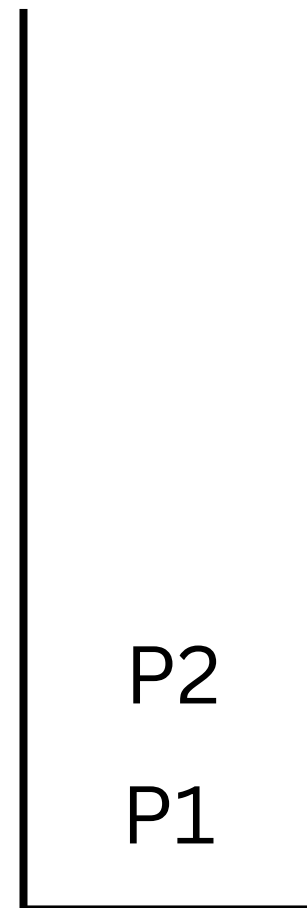
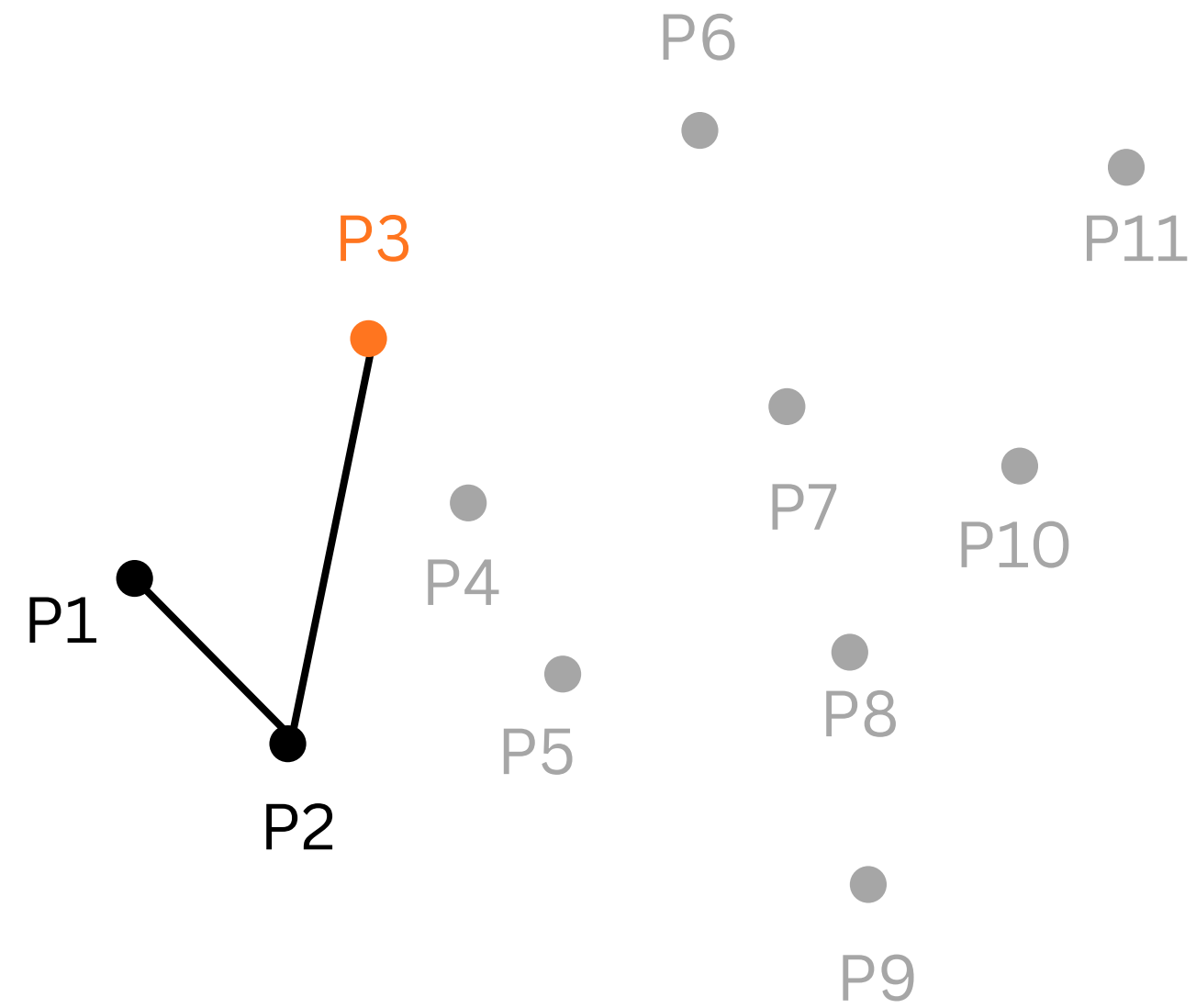
P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 P11



Stack

Array

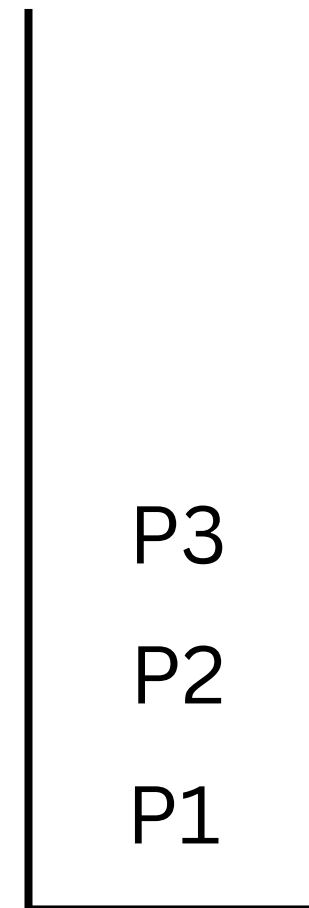
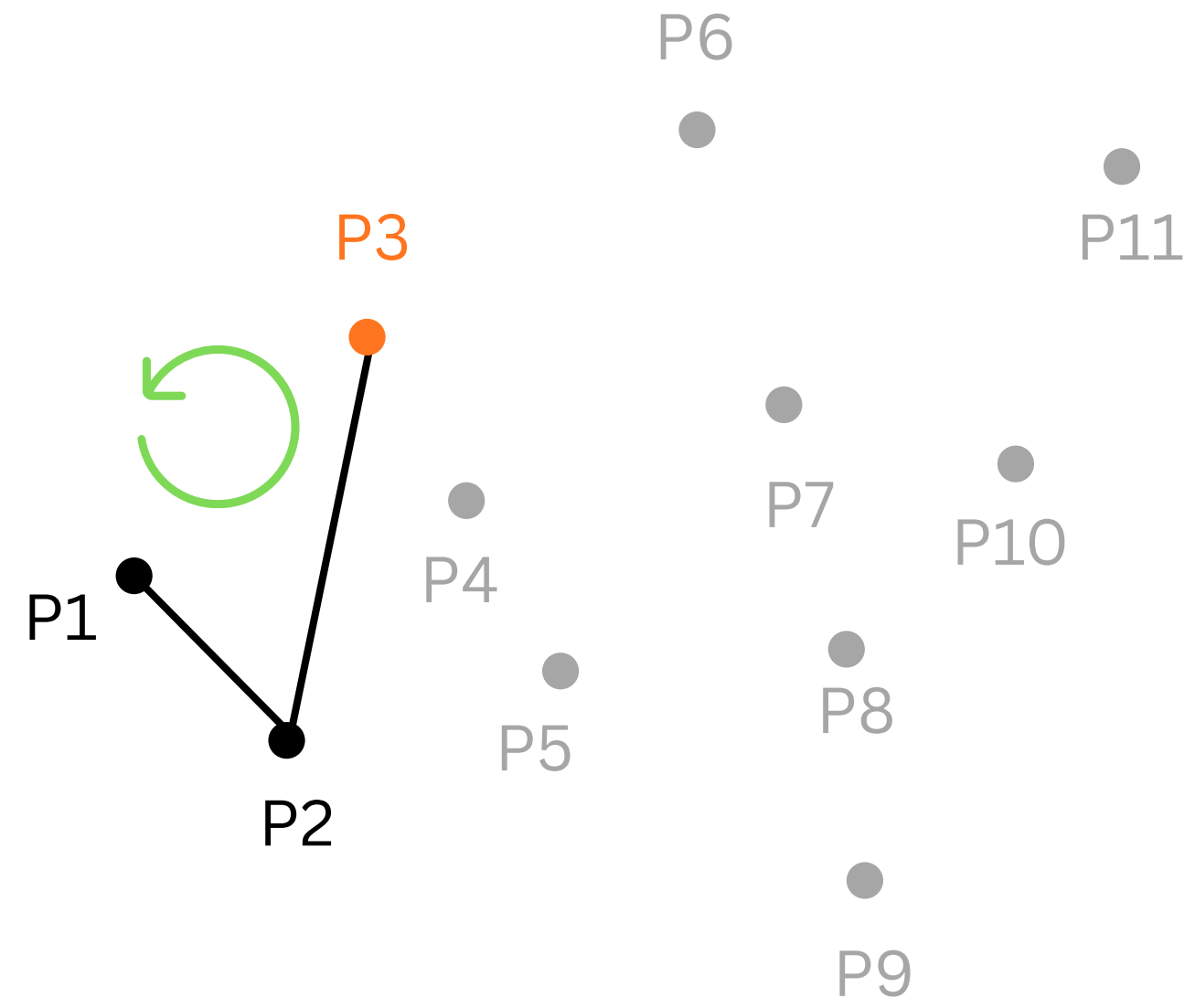
P1 P2 **P3** P4 P5 P6 P7 P8 P9 P10 P11



Stack

Array

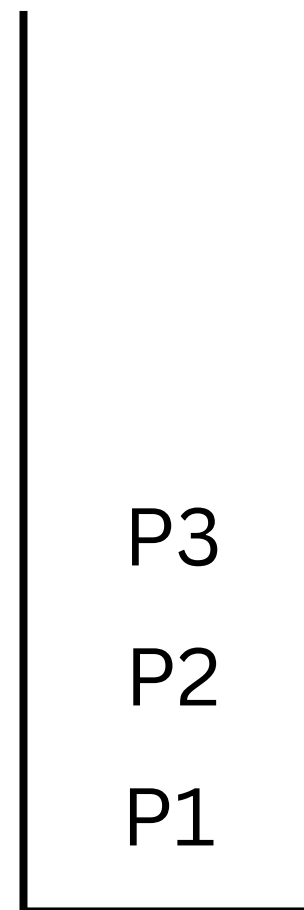
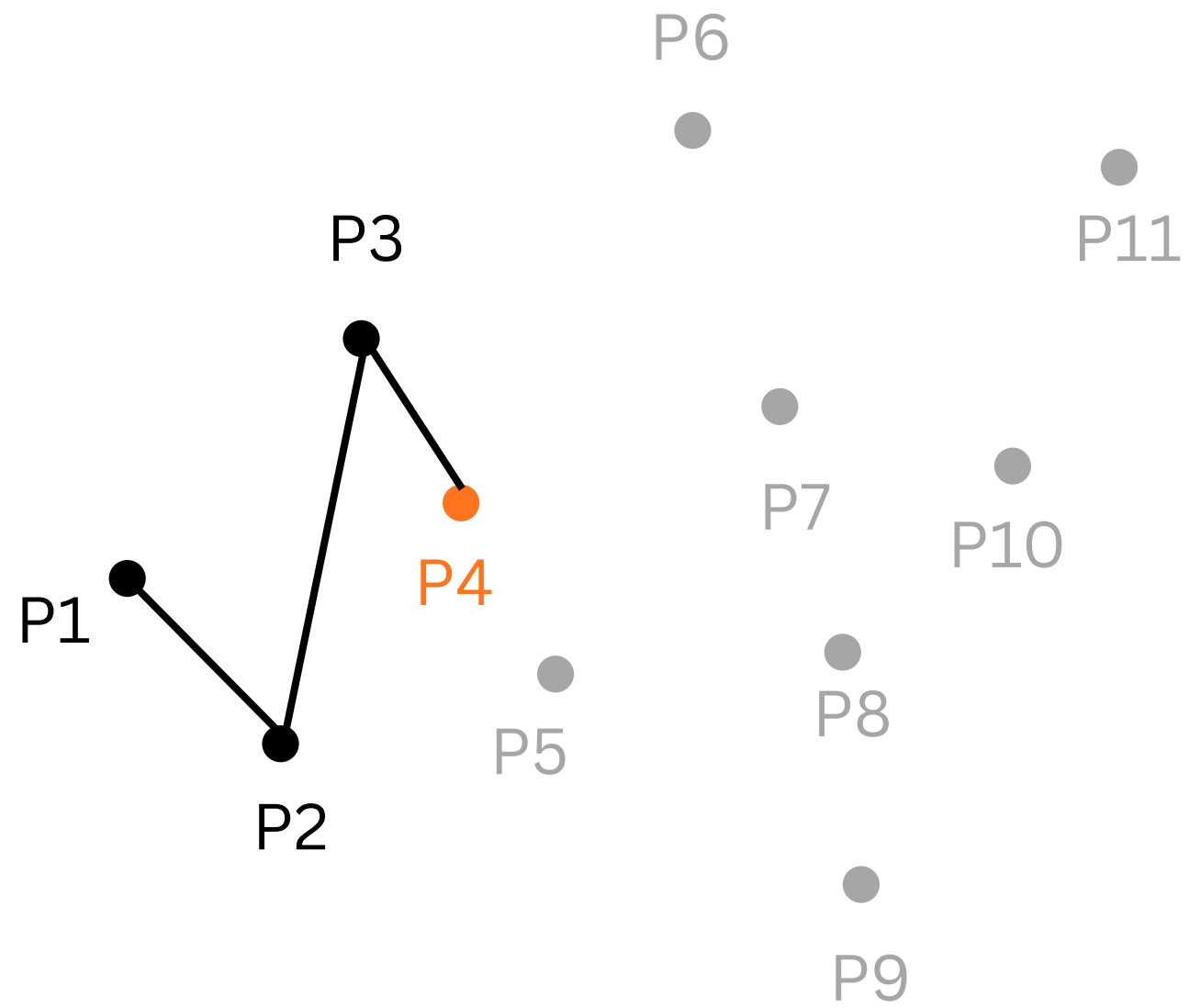
P1 P2 **P3** P4 P5 P6 P7 P8 P9 P10 P11



Stack

Array

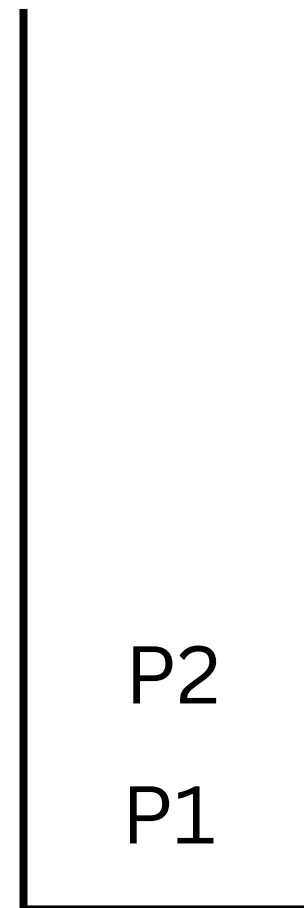
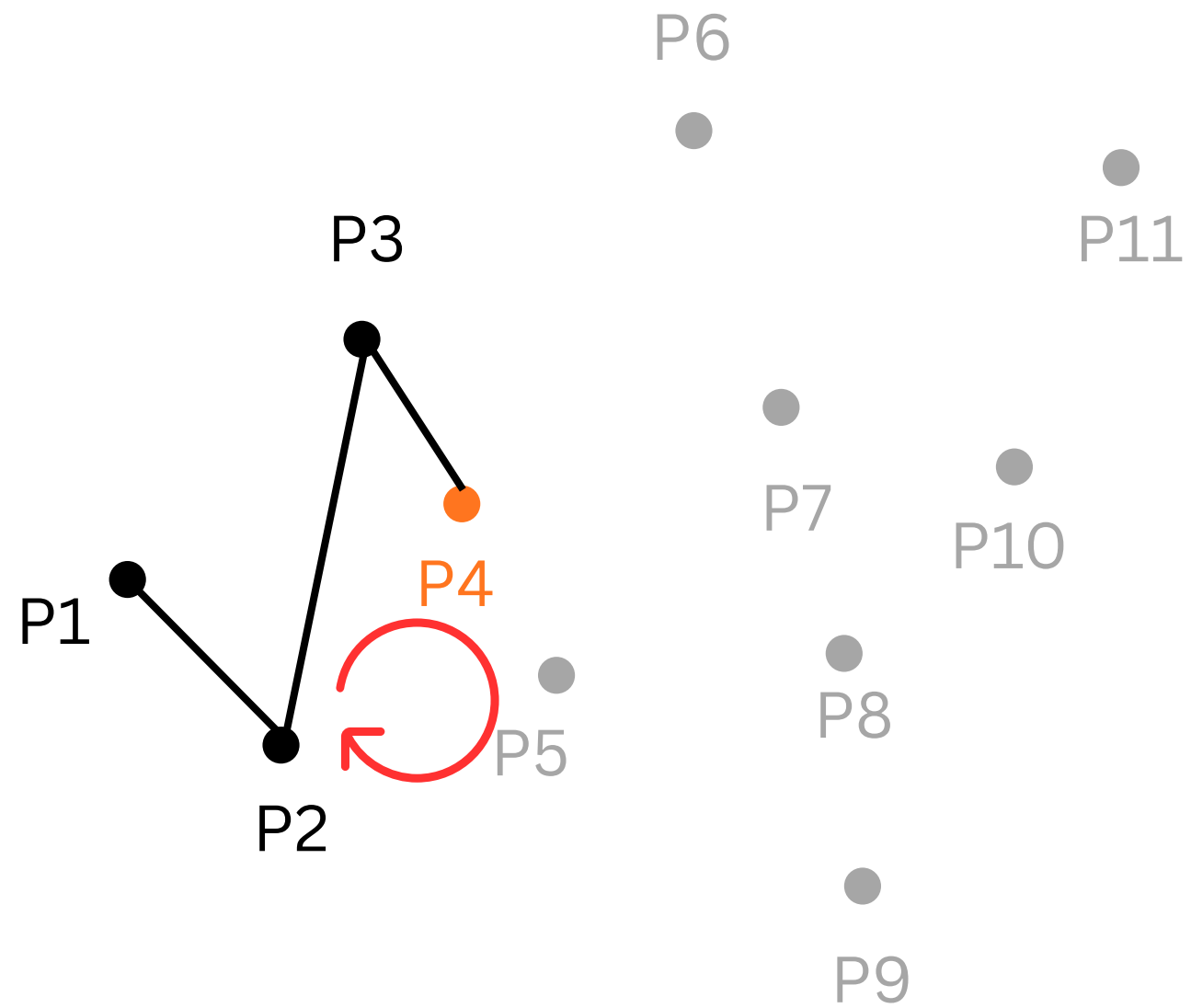
P1 P2 P3 **P4** P5 P6 P7 P8 P9 P10 P11



Stack

Array

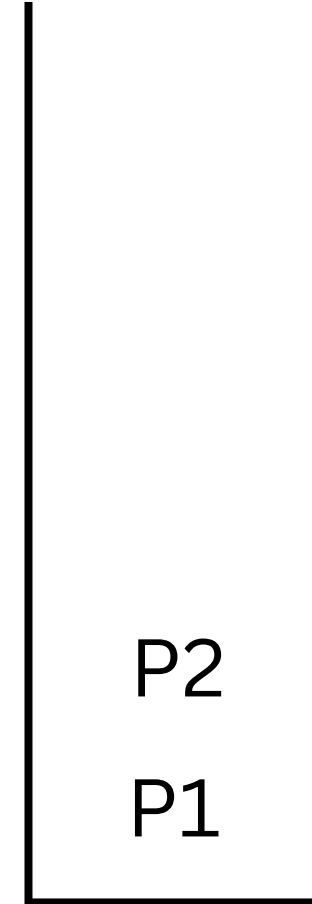
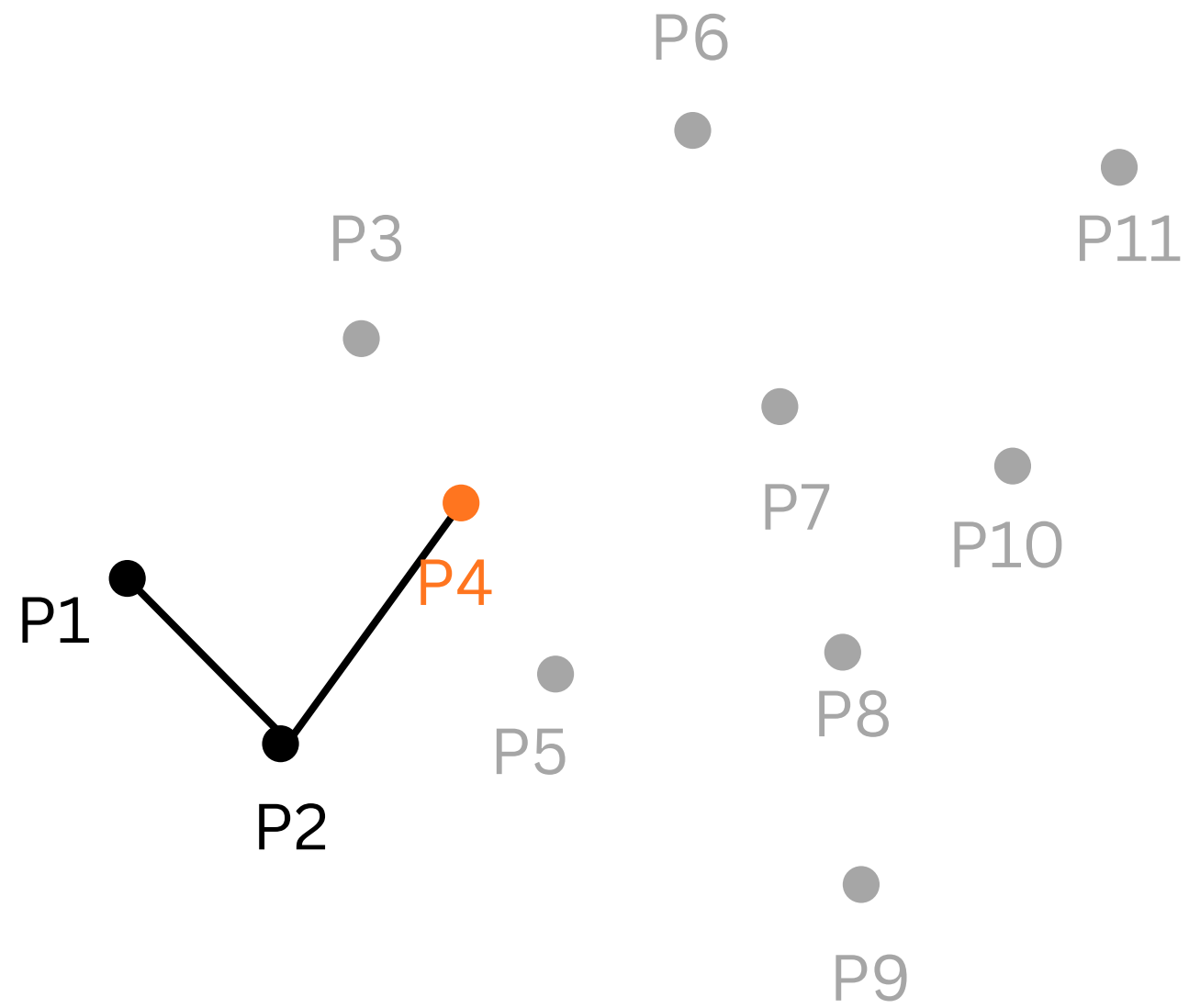
P1 P2 P3 **P4** P5 P6 P7 P8 P9 P10 P11



Stack

Array

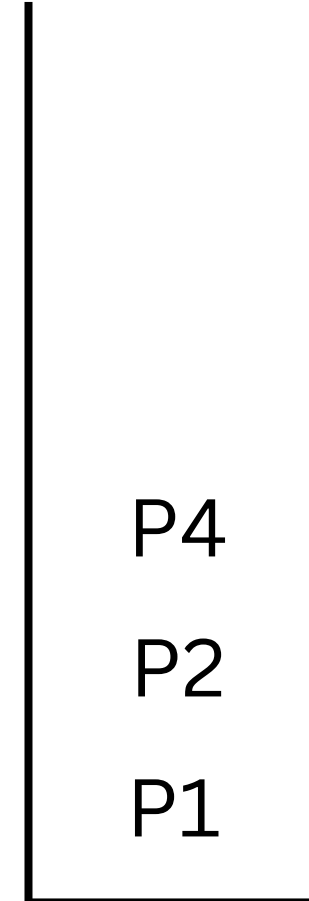
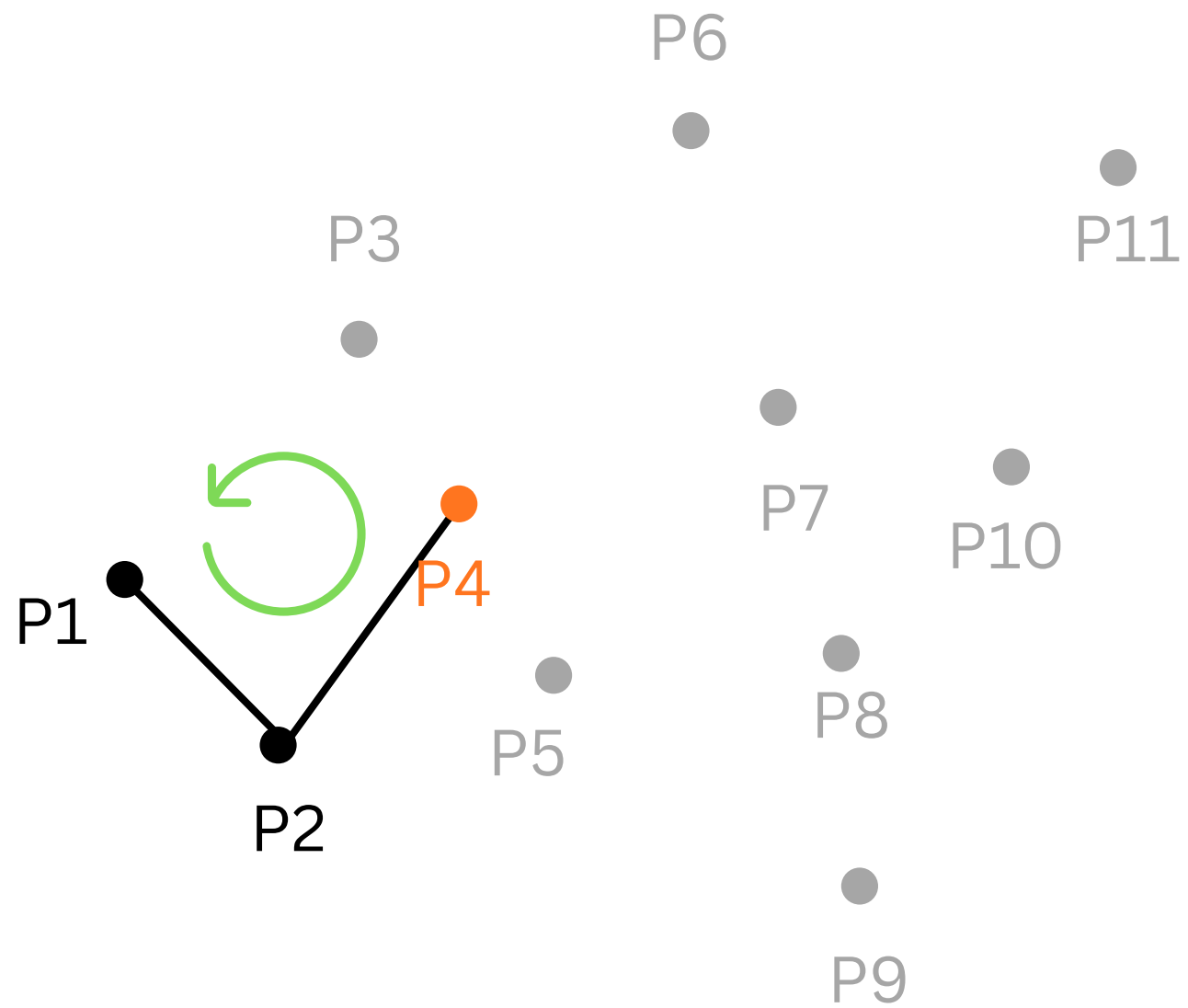
P1 P2 P3 **P4** P5 P6 P7 P8 P9 P10 P11



Stack

Array

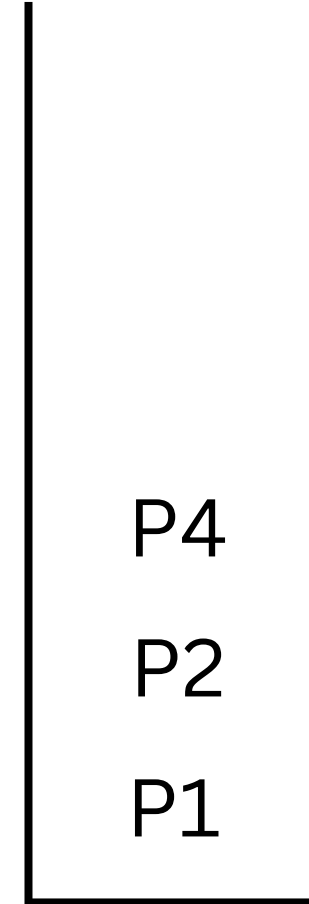
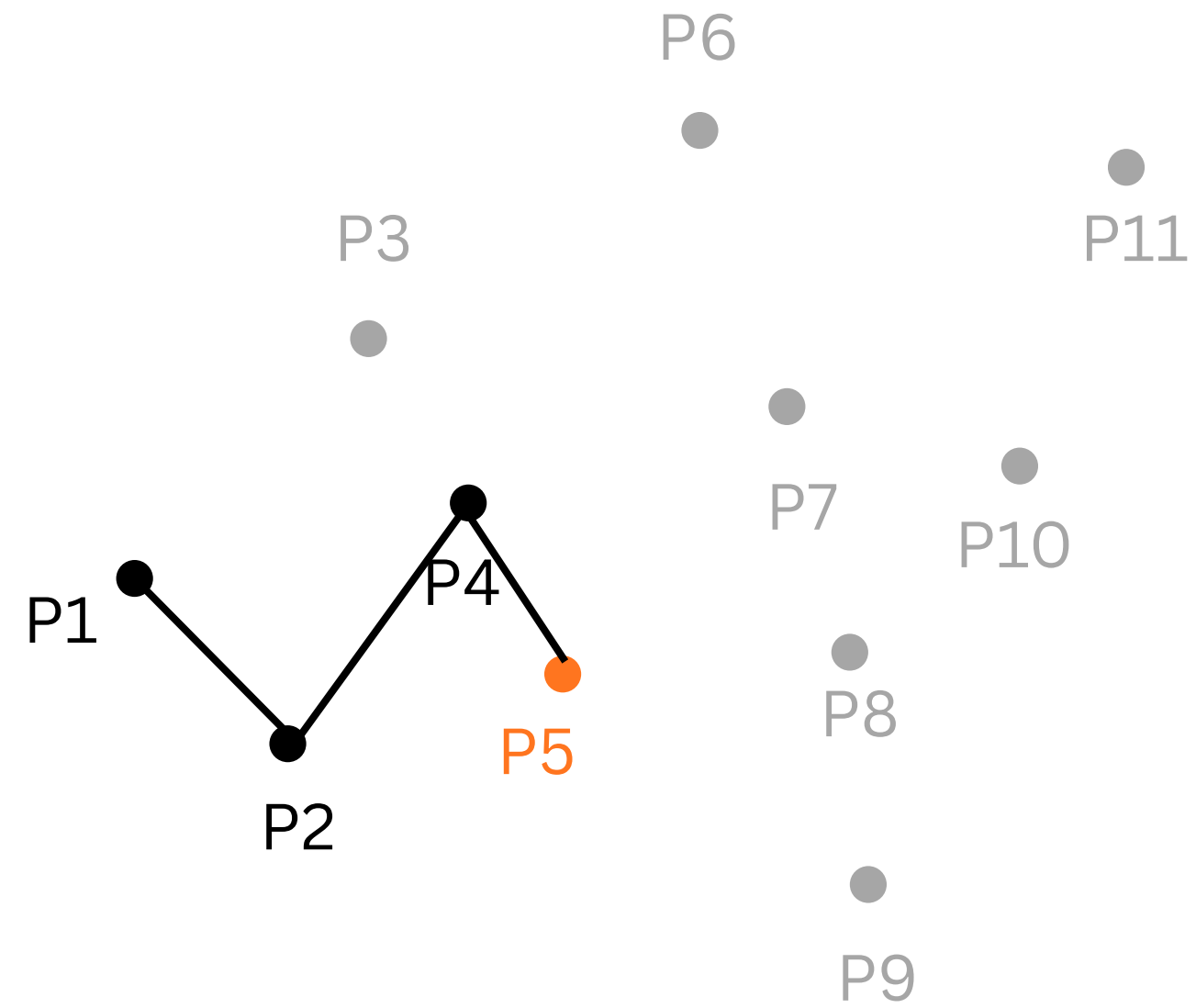
P1 P2 P3 **P4** P5 P6 P7 P8 P9 P10 P11



Stack

Array

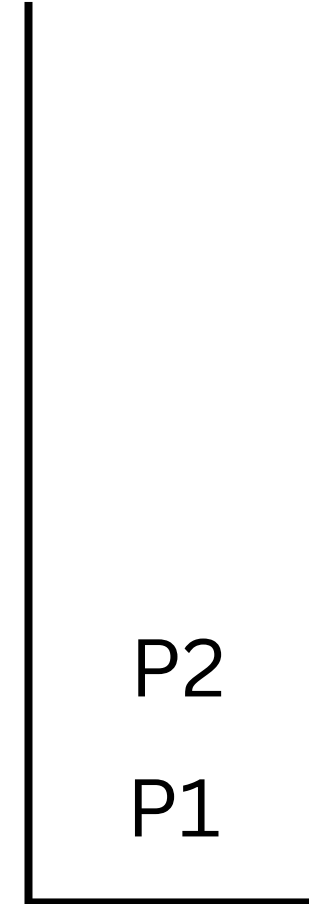
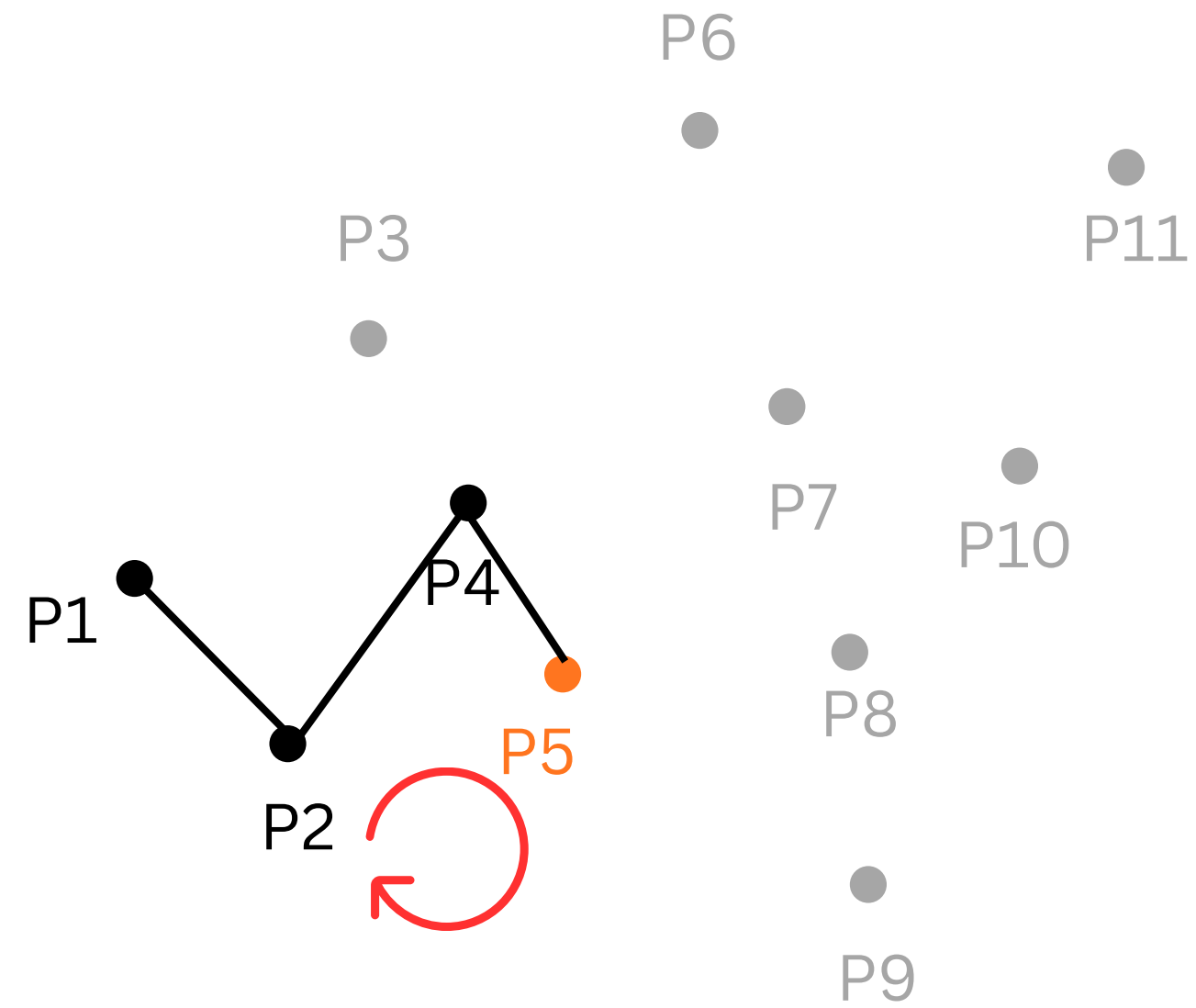
P1 P2 P3 P4 **P5** P6 P7 P8 P9 P10 P11



Stack

Array

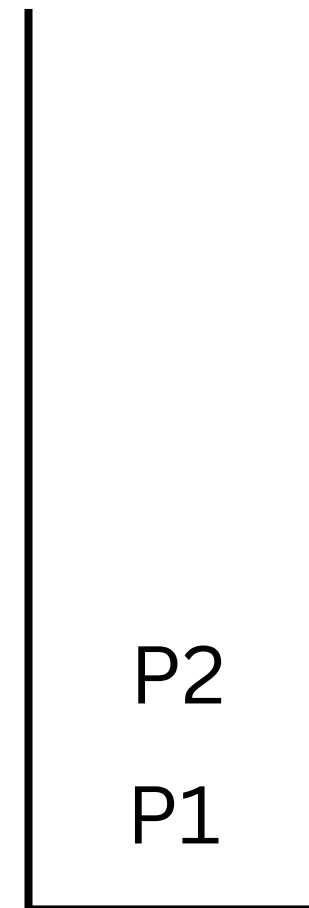
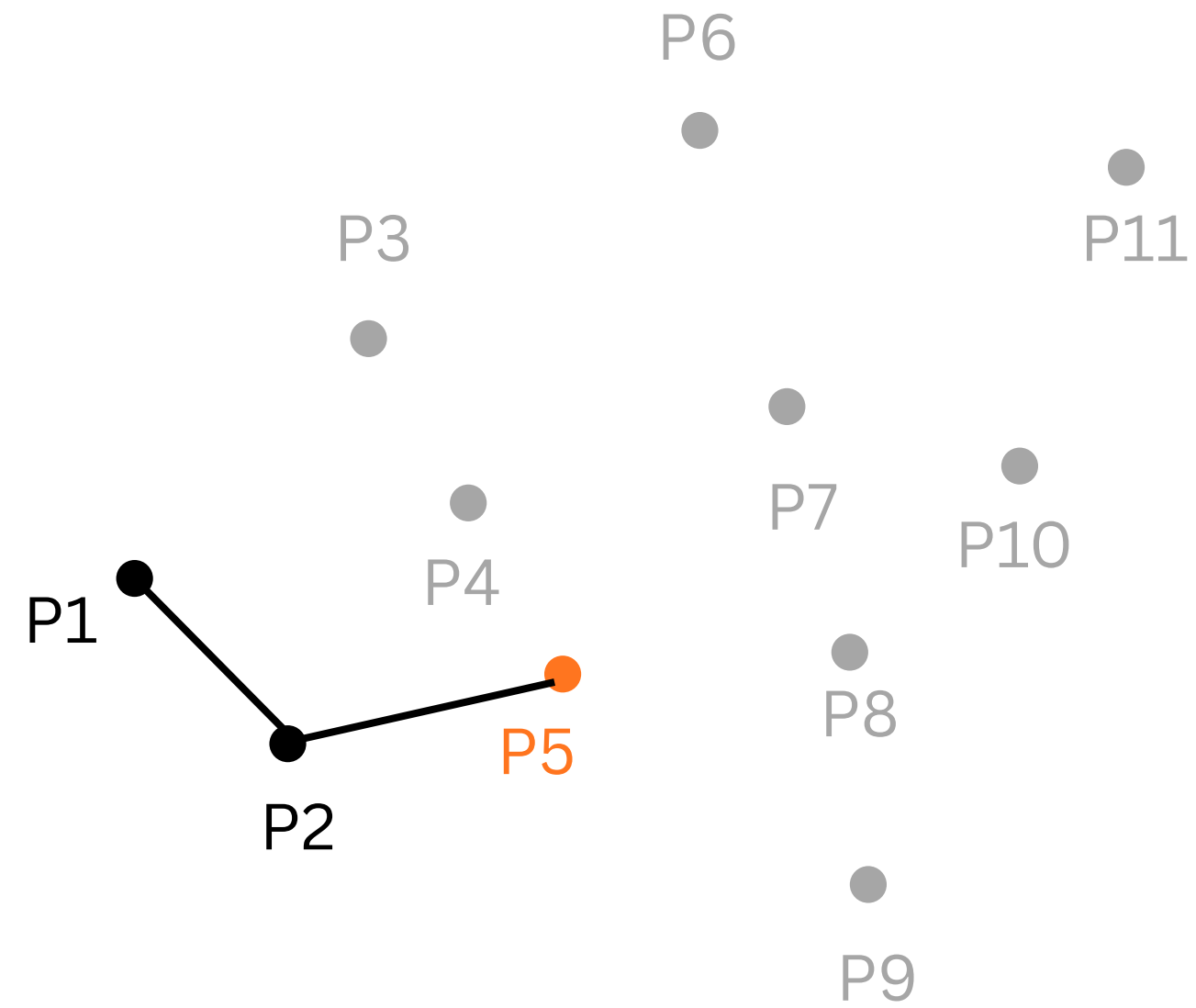
P1 P2 P3 P4 **P5** P6 P7 P8 P9 P10 P11



Stack

Array

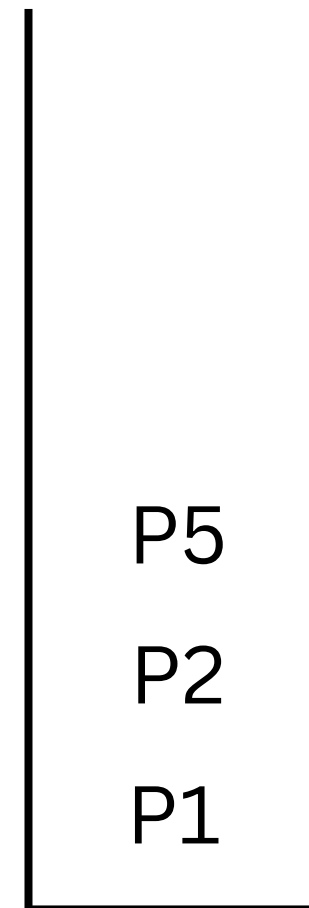
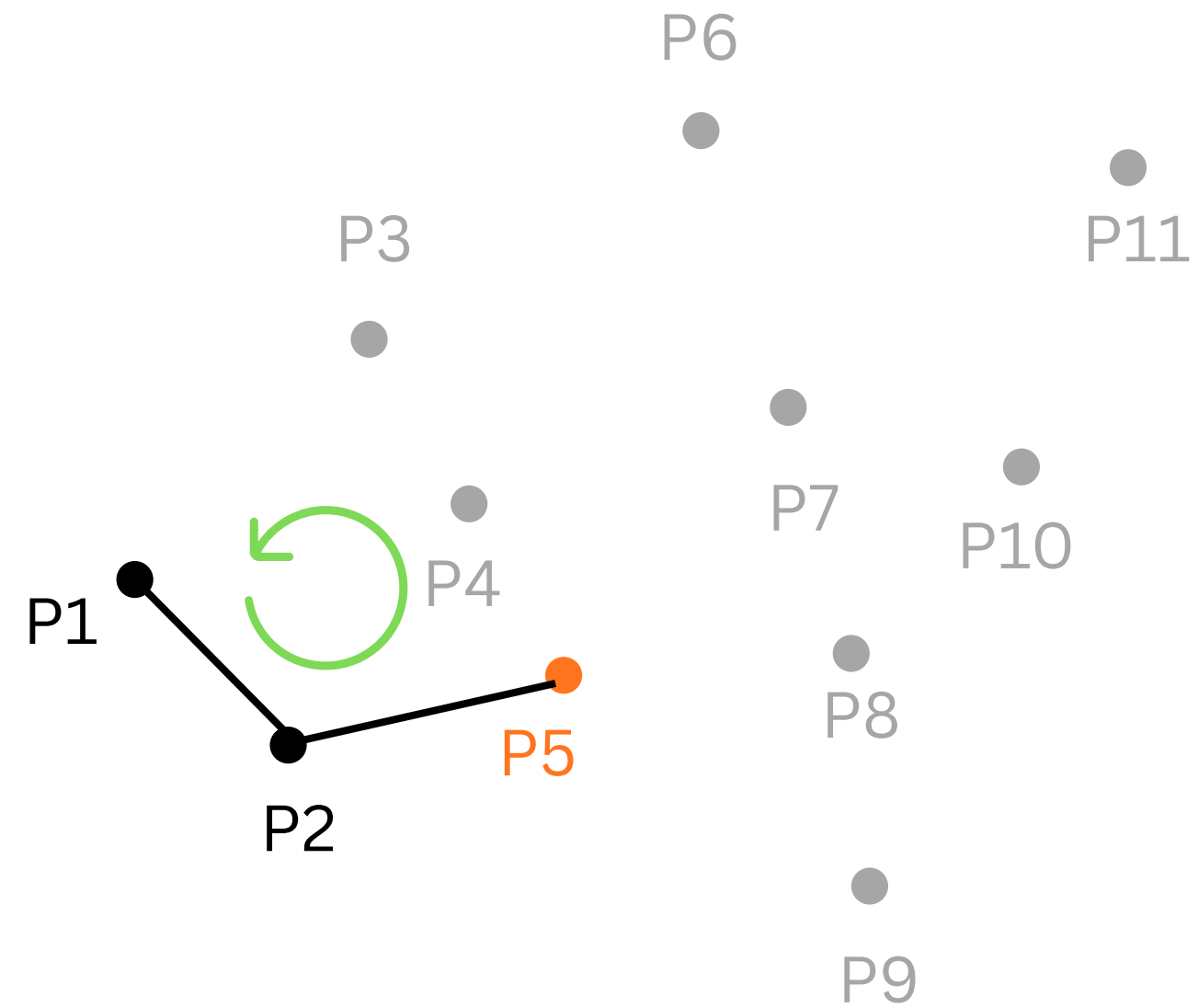
P1 P2 P3 P4 **P5** P6 P7 P8 P9 P10 P11



Stack

Array

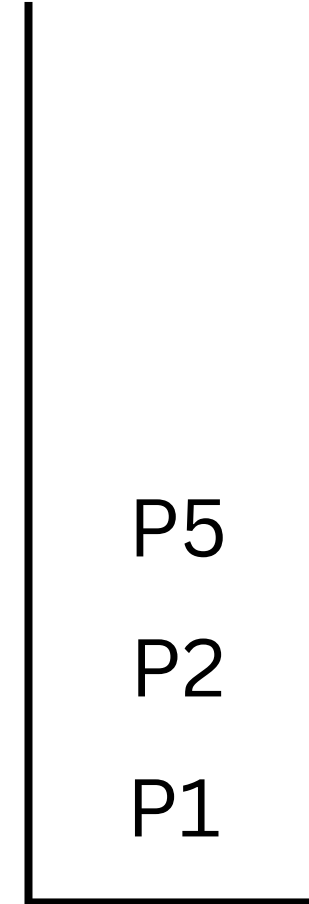
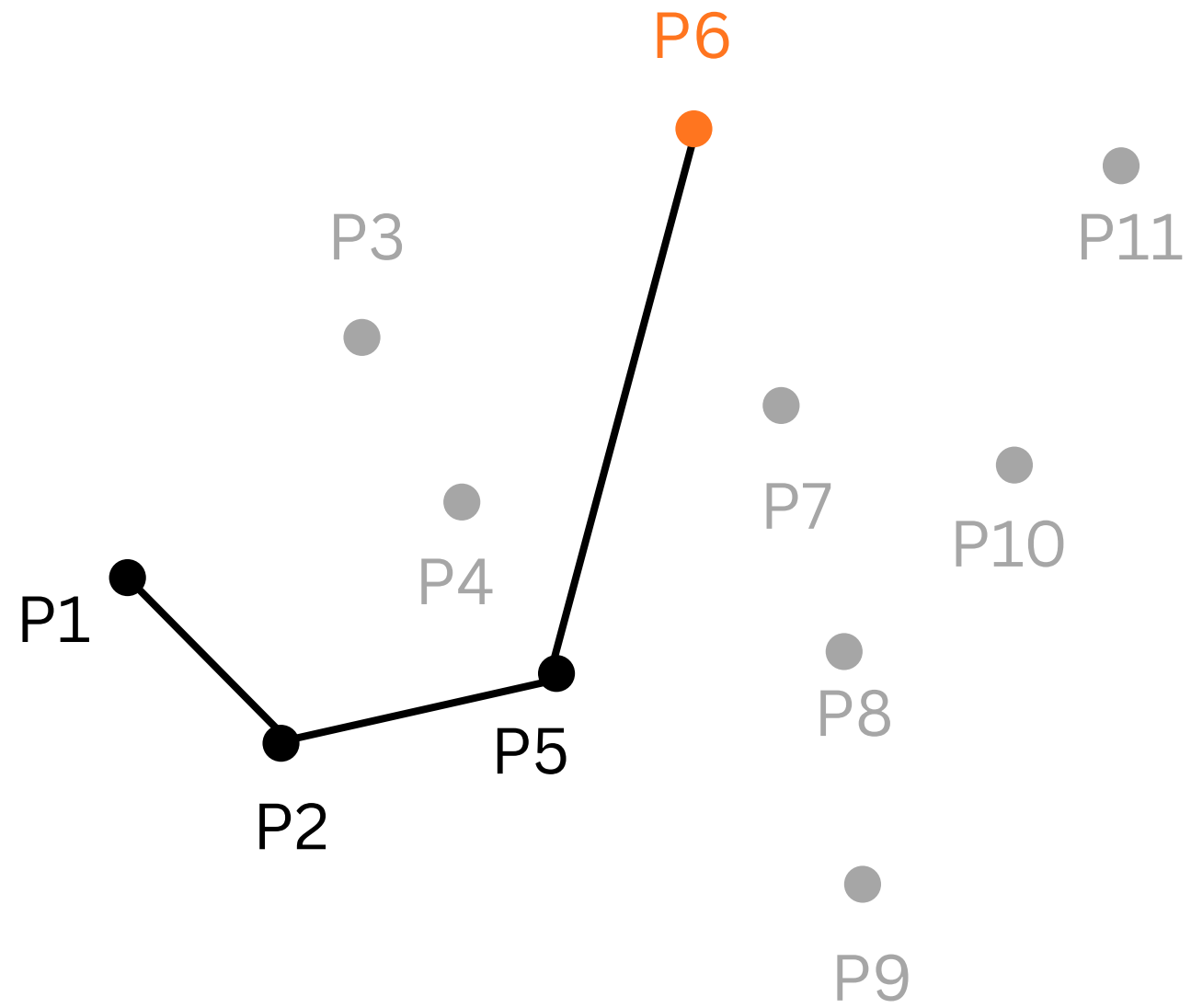
P1 P2 P3 P4 **P5** P6 P7 P8 P9 P10 P11



Stack

Array

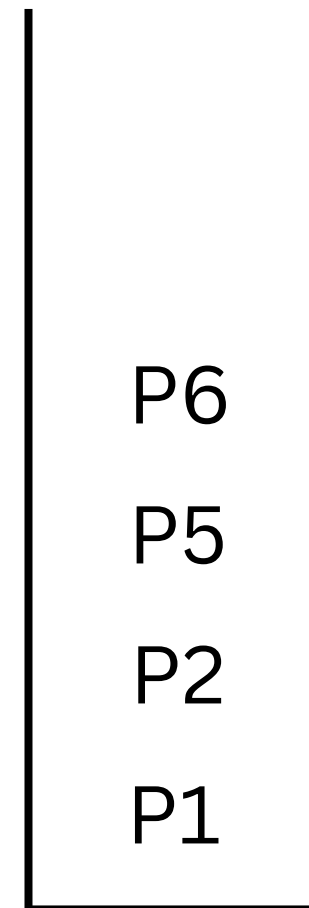
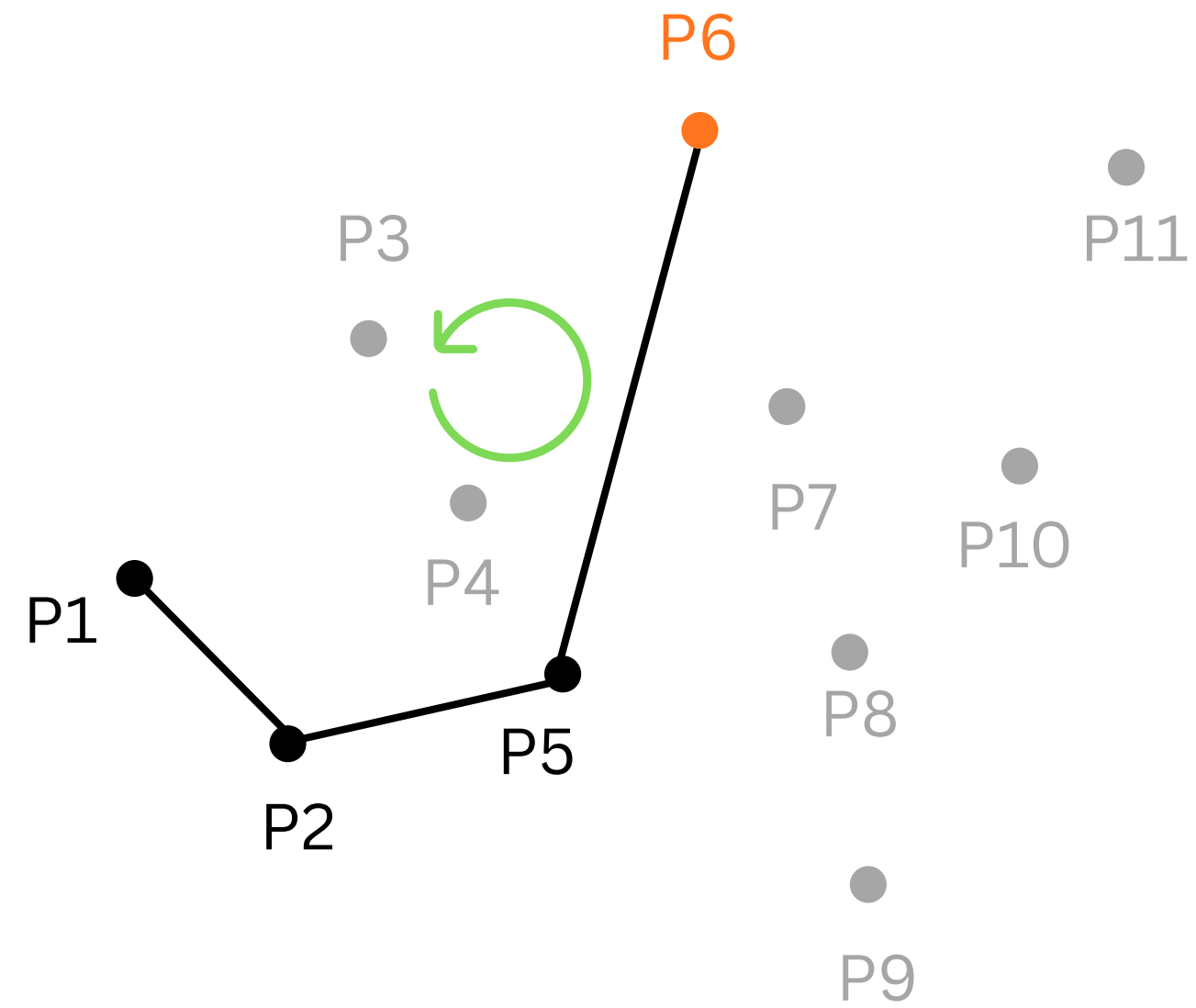
P1 P2 P3 P4 P5 **P6** P7 P8 P9 P10 P11



Stack

Array

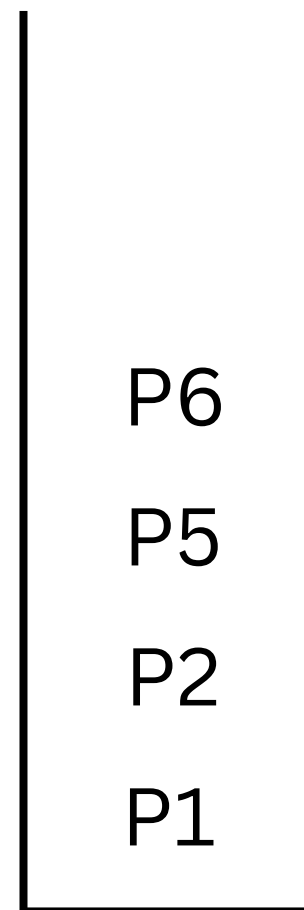
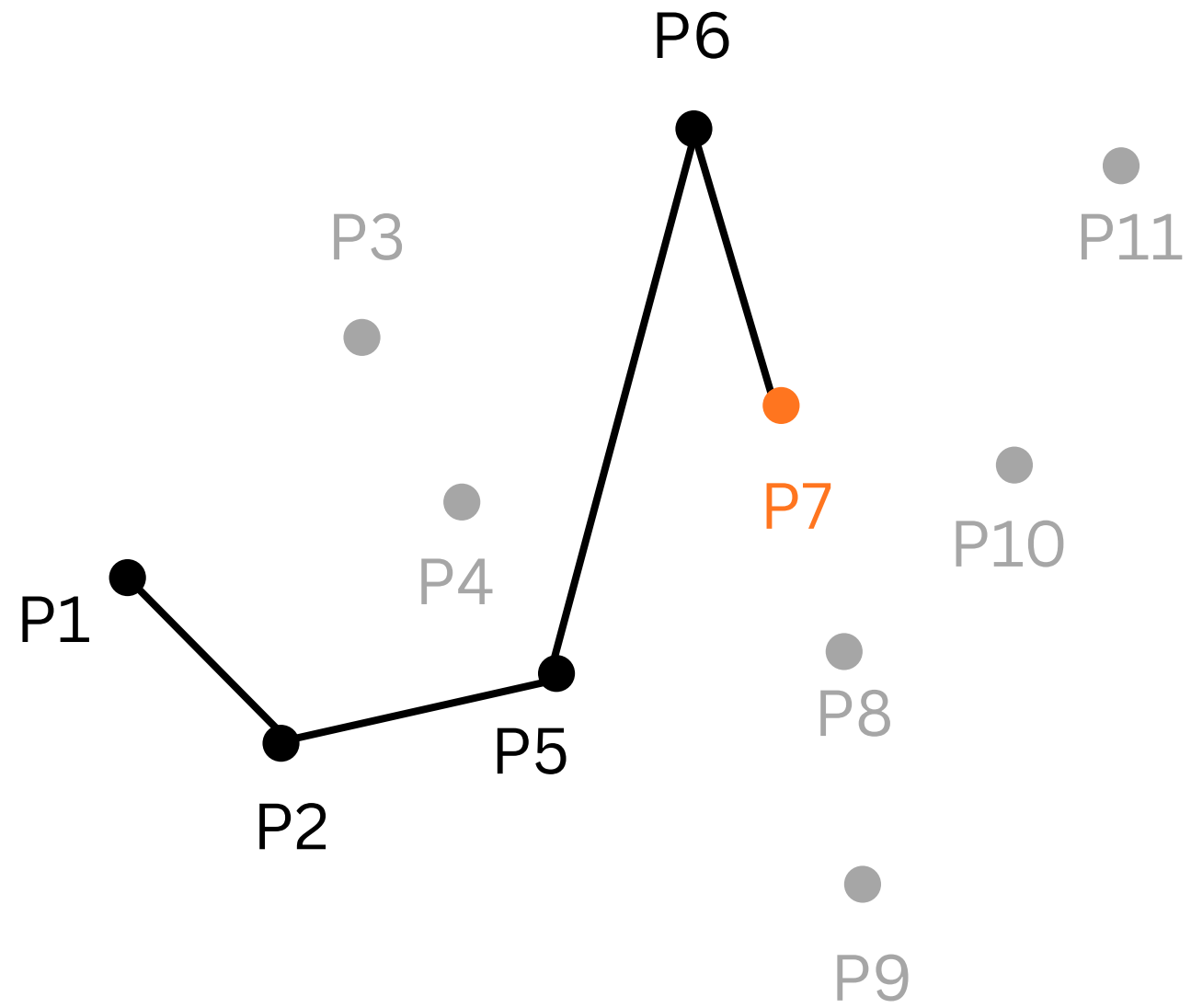
P1 P2 P3 P4 P5 **P6** P7 P8 P9 P10 P11



Stack

Array

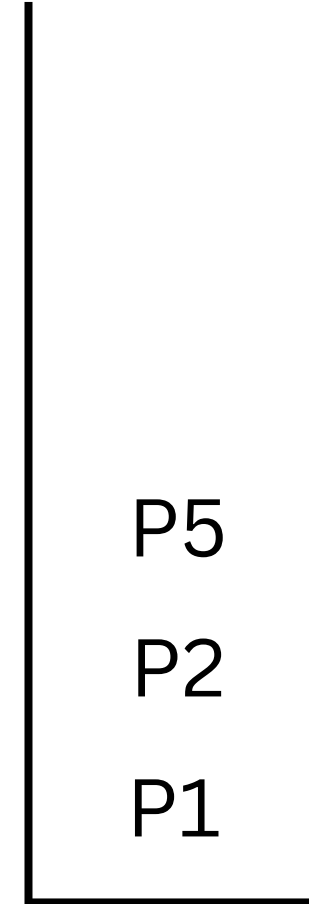
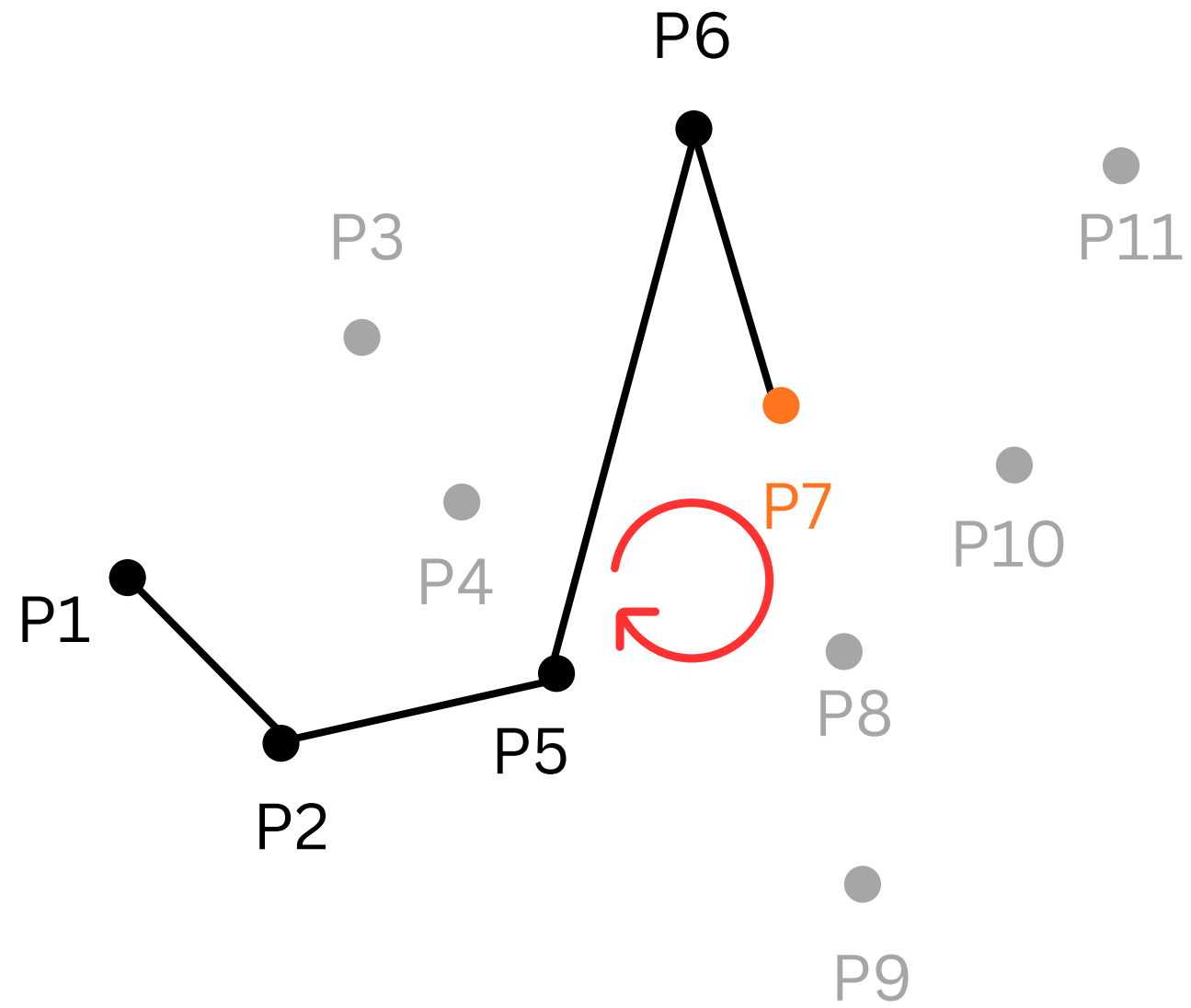
P1 P2 P3 P4 P5 P6 **P7** P8 P9 P10 P11



Stack

Array

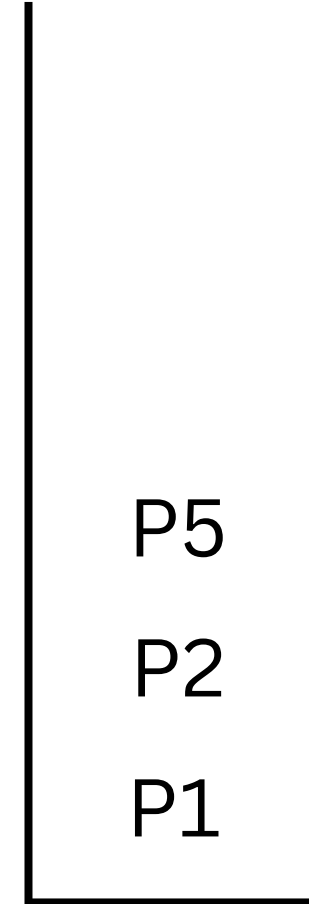
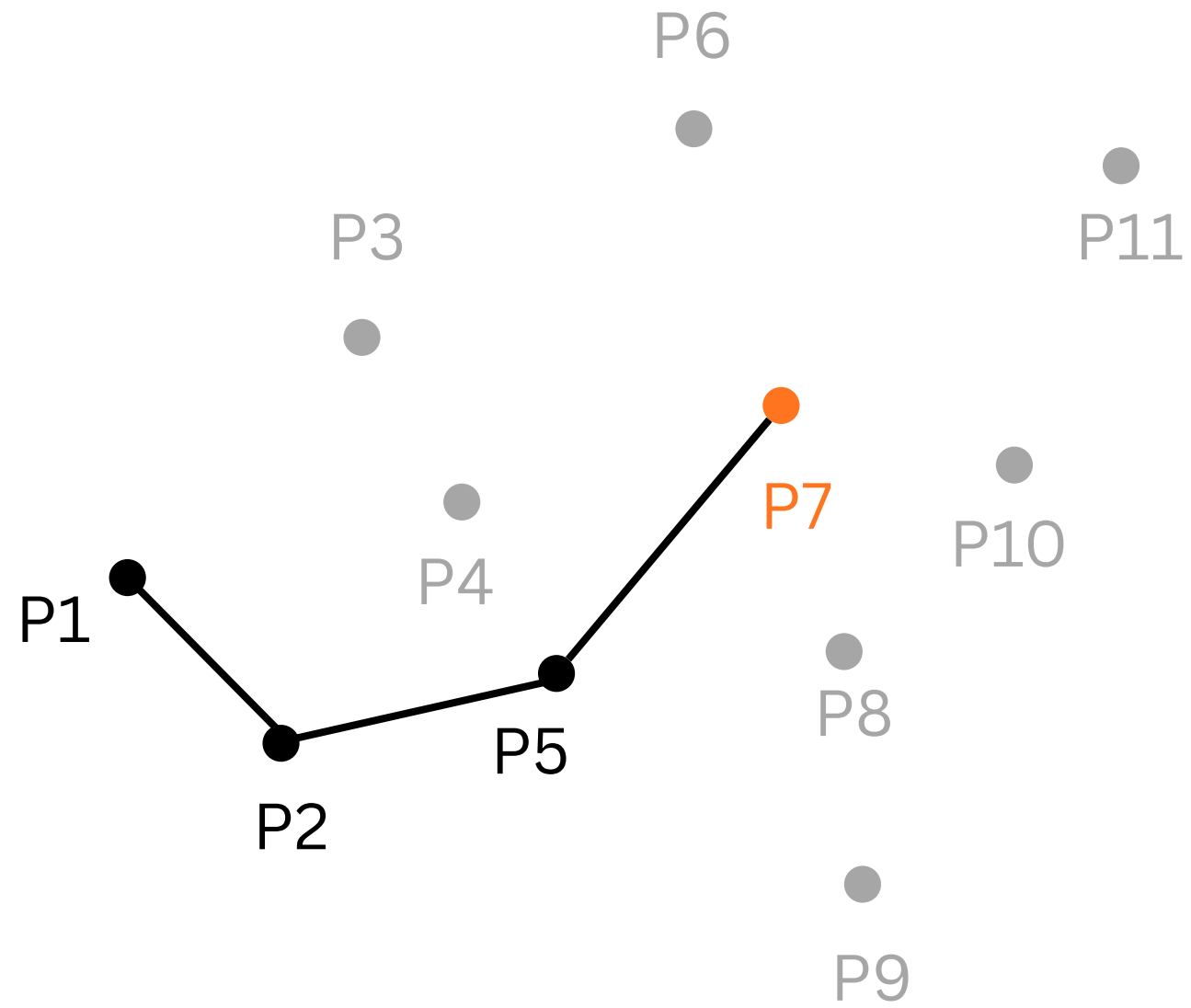
P1 P2 P3 P4 P5 P6 **P7** P8 P9 P10 P11



Stack

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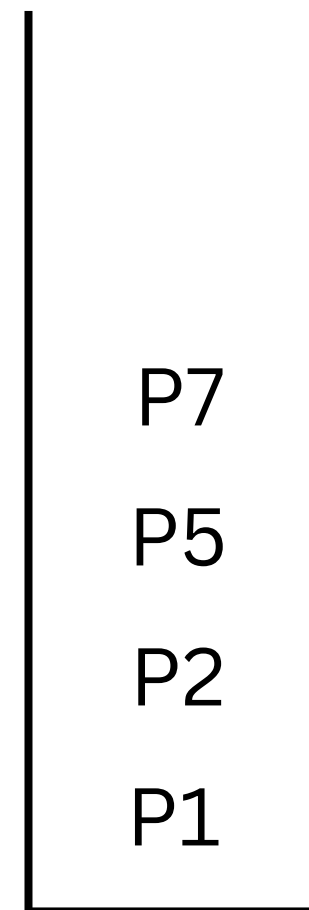
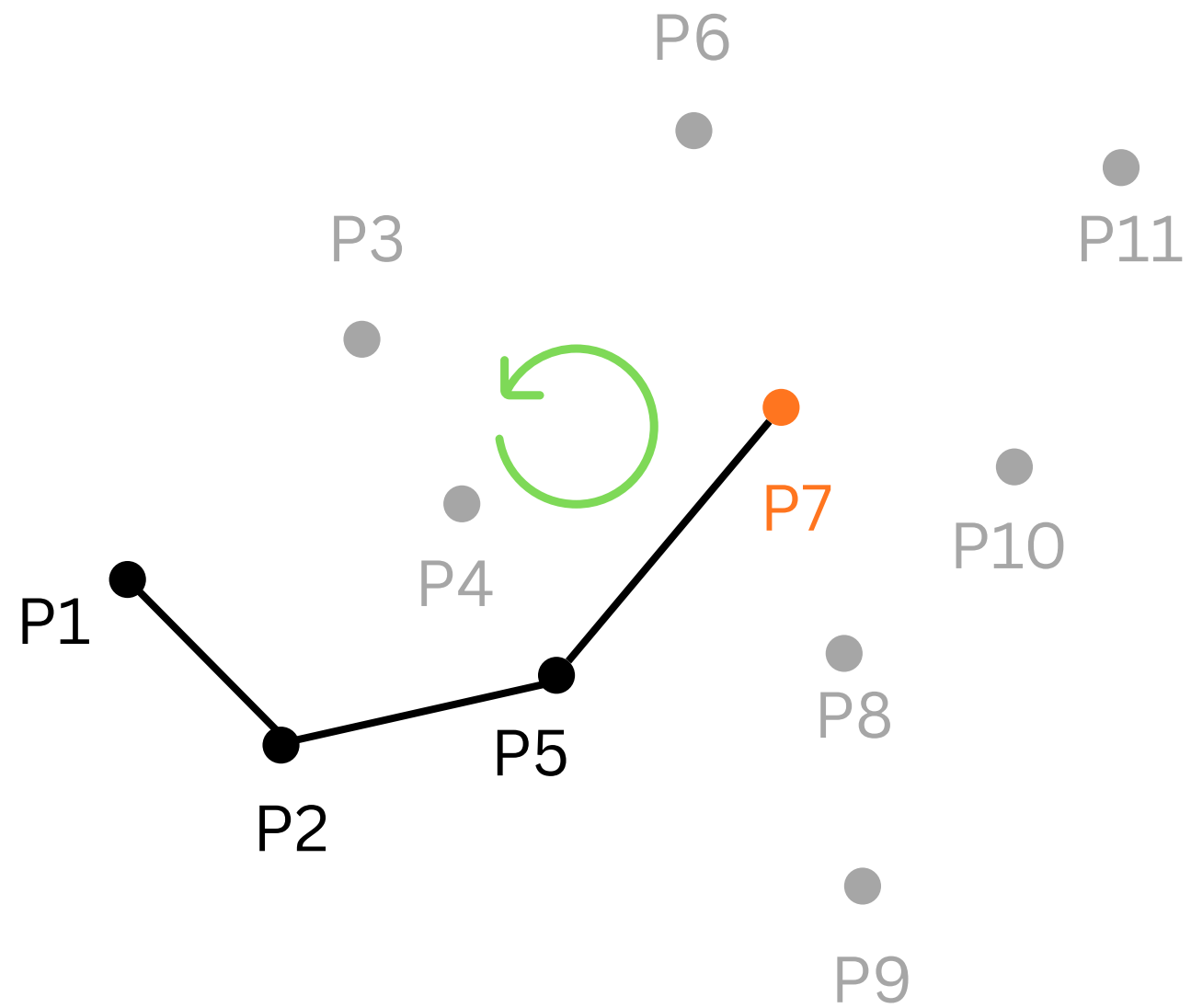
P1 P2 P3 P4 P5 P6 **P7** P8 P9 P10 P11



Stack

Array

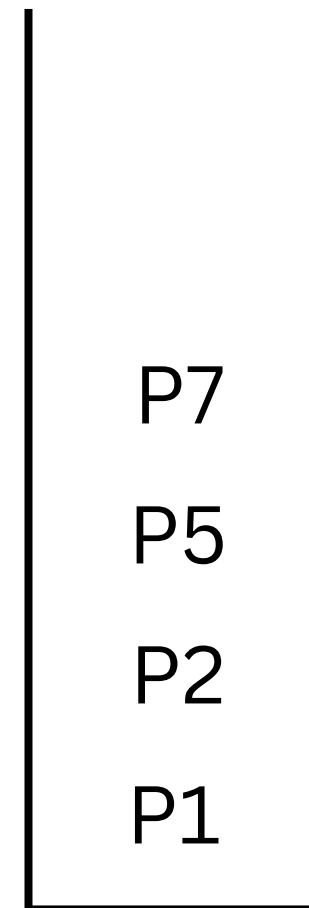
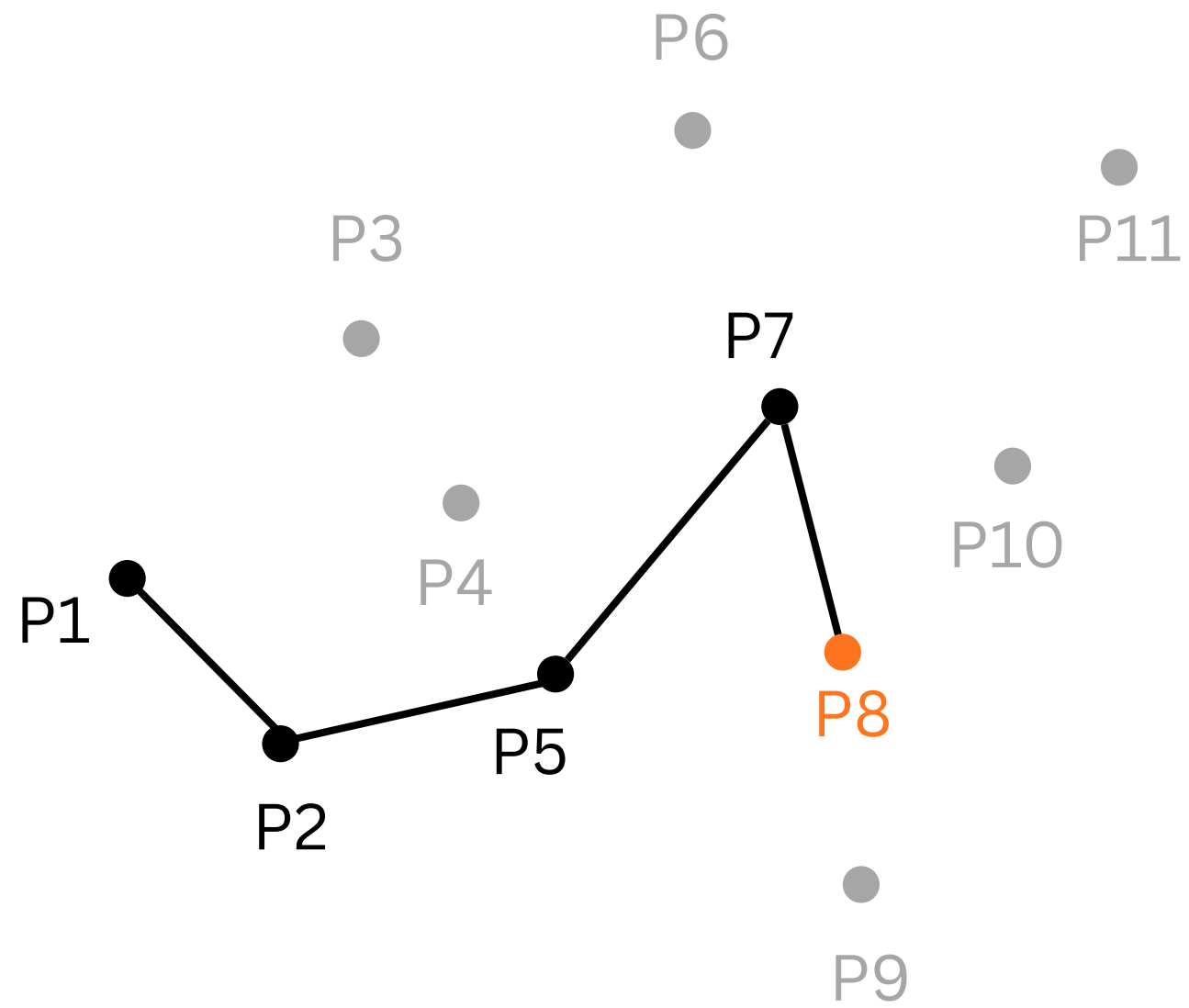
P1 P2 P3 P4 P5 P6 **P7** P8 P9 P10 P11



Stack

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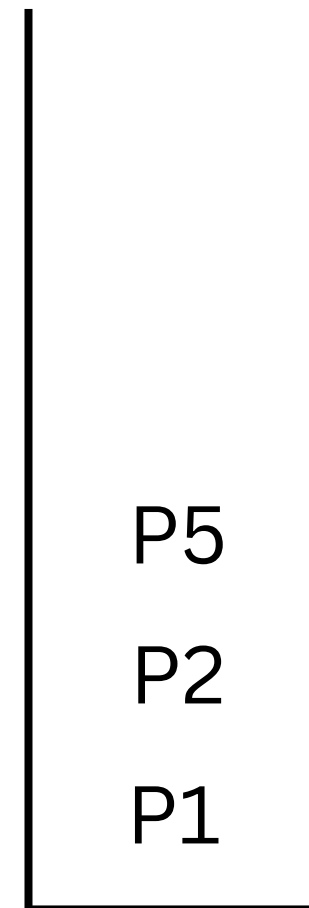
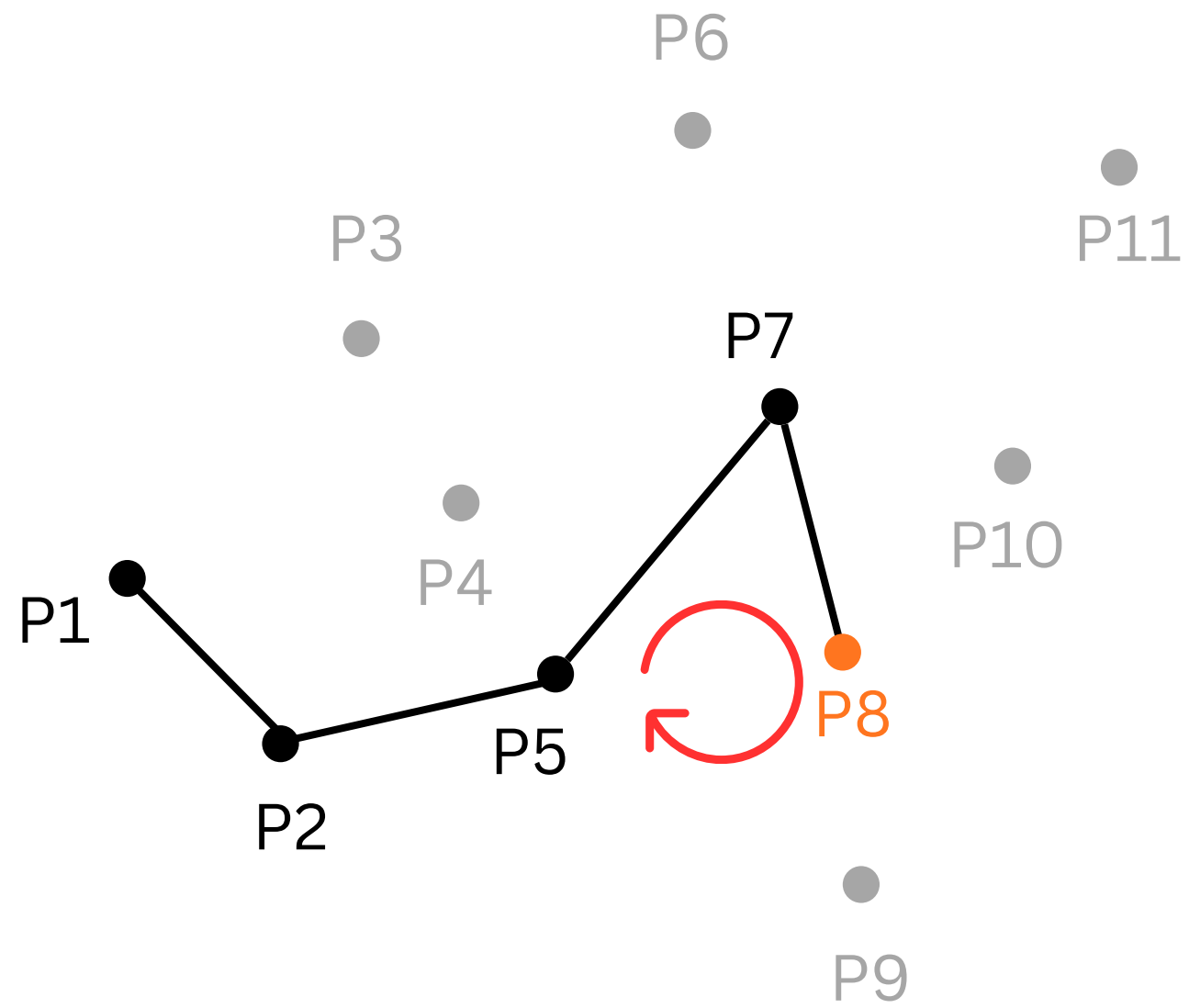
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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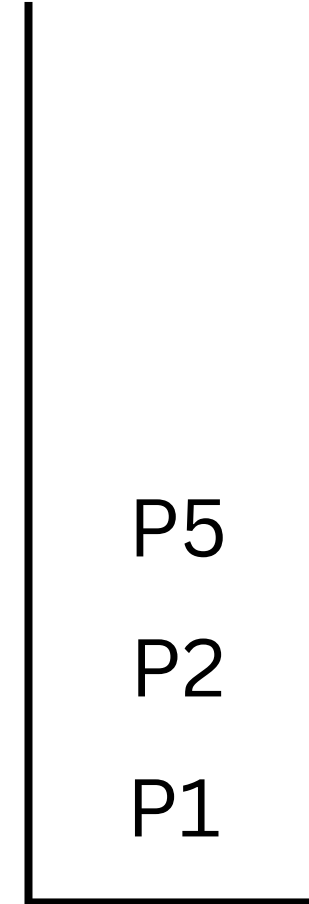
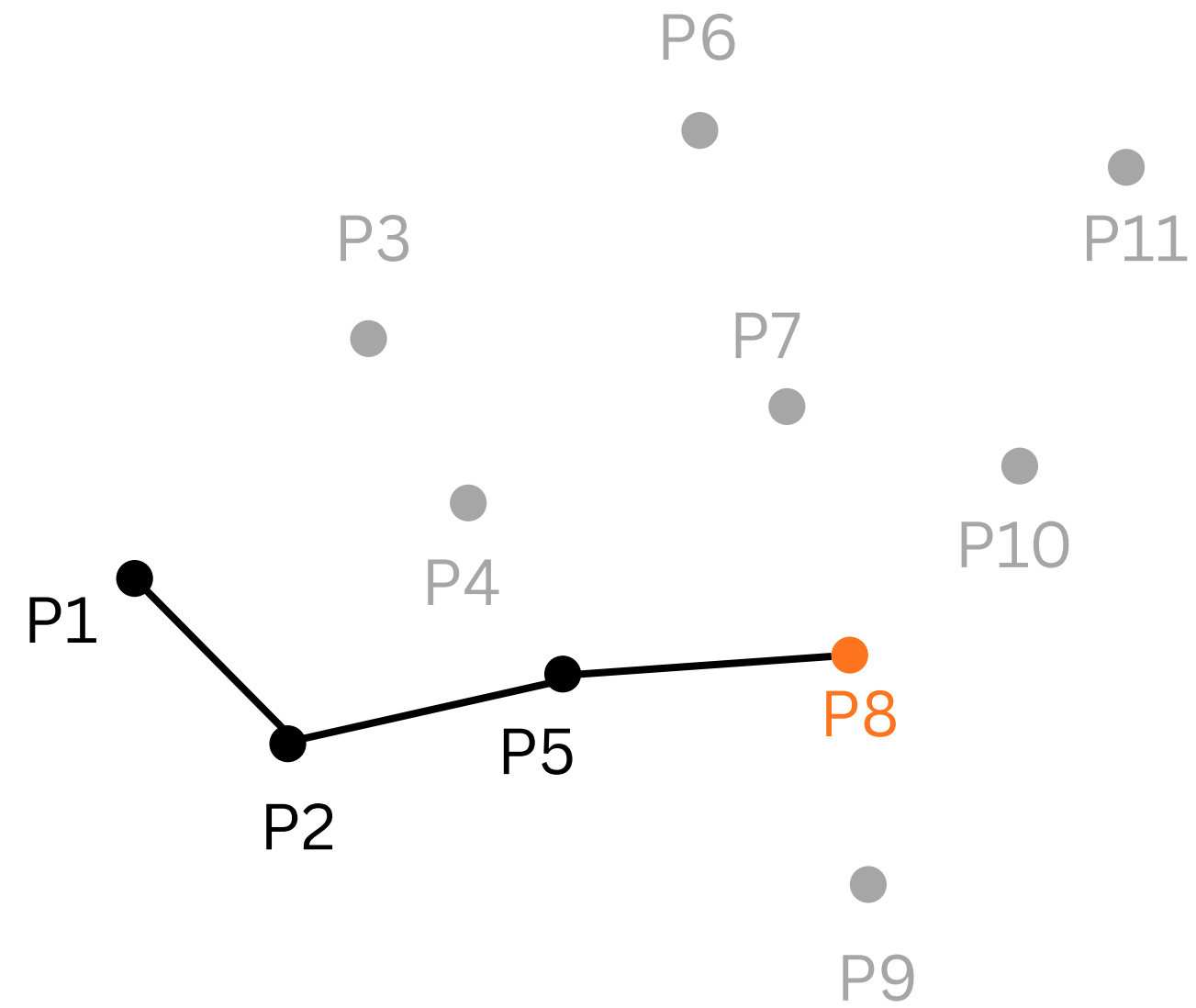
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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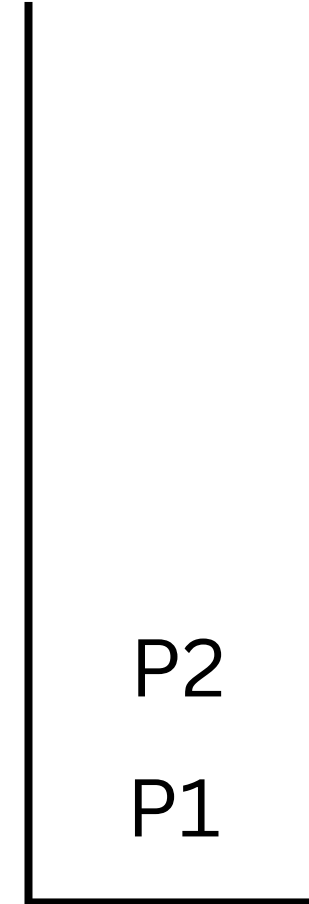
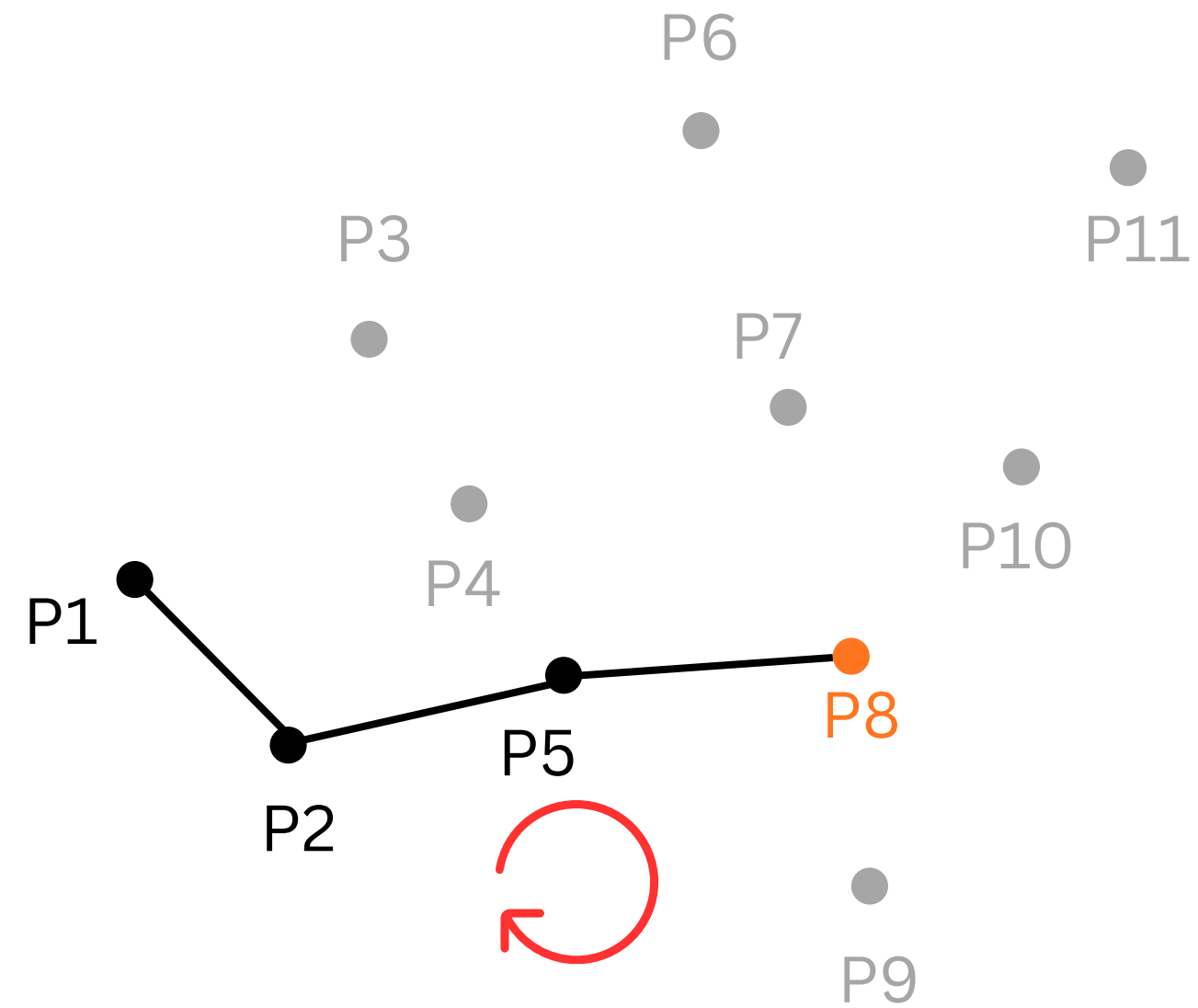
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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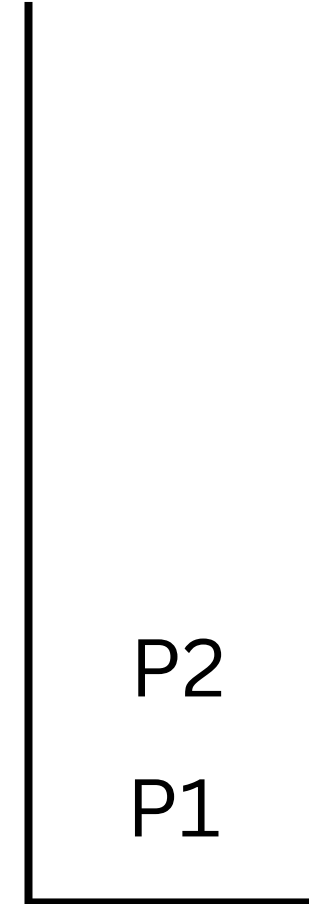
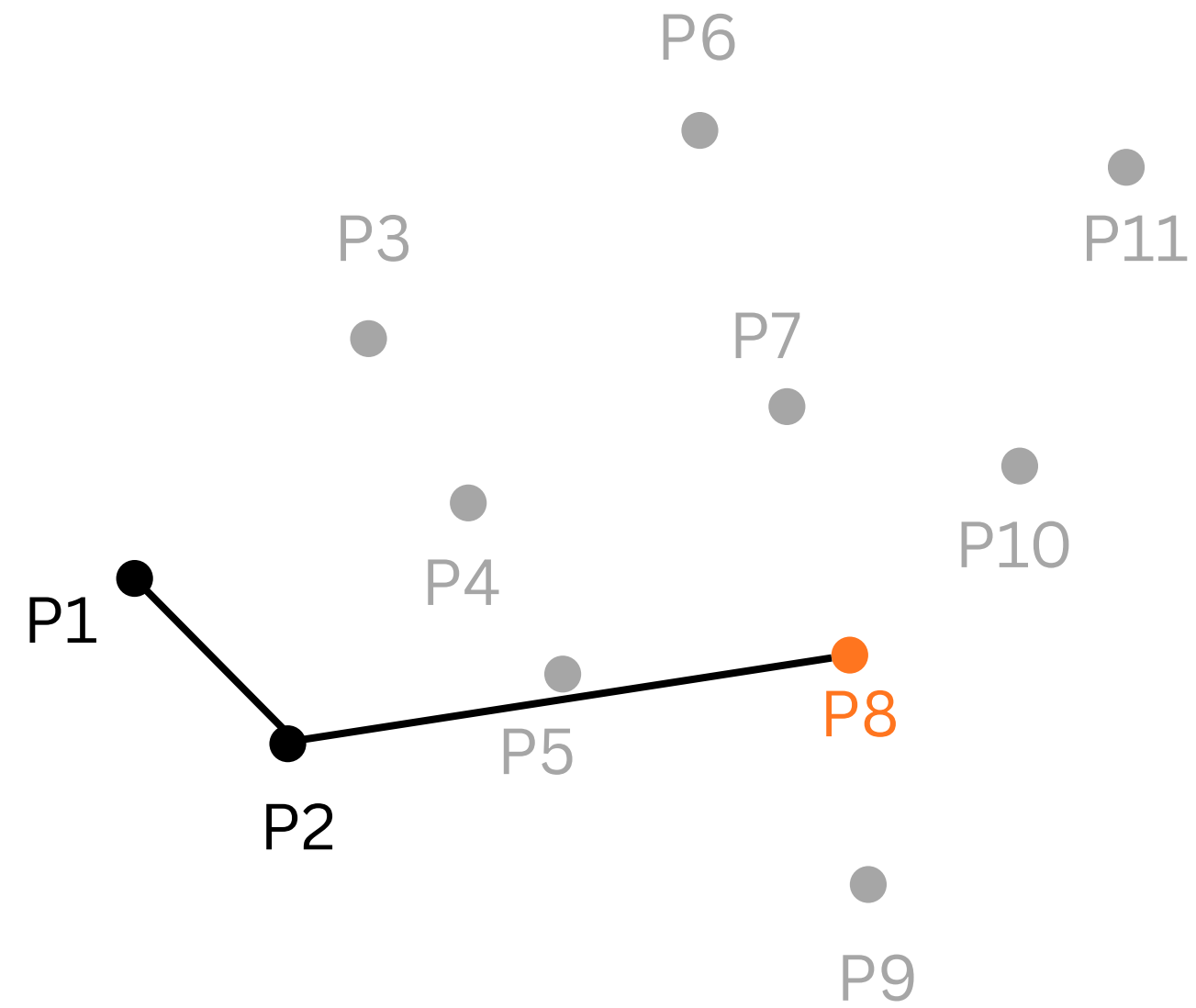
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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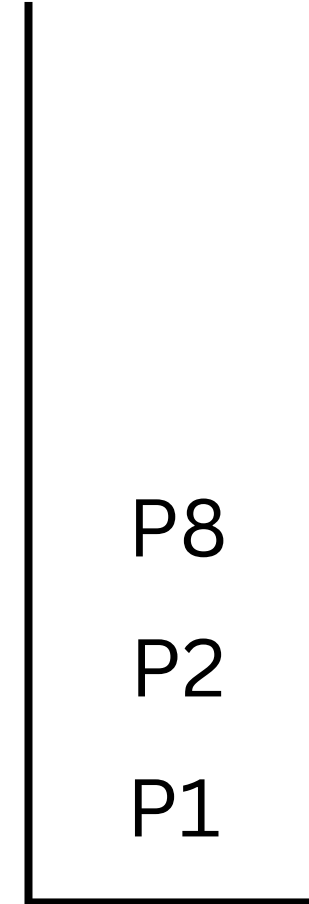
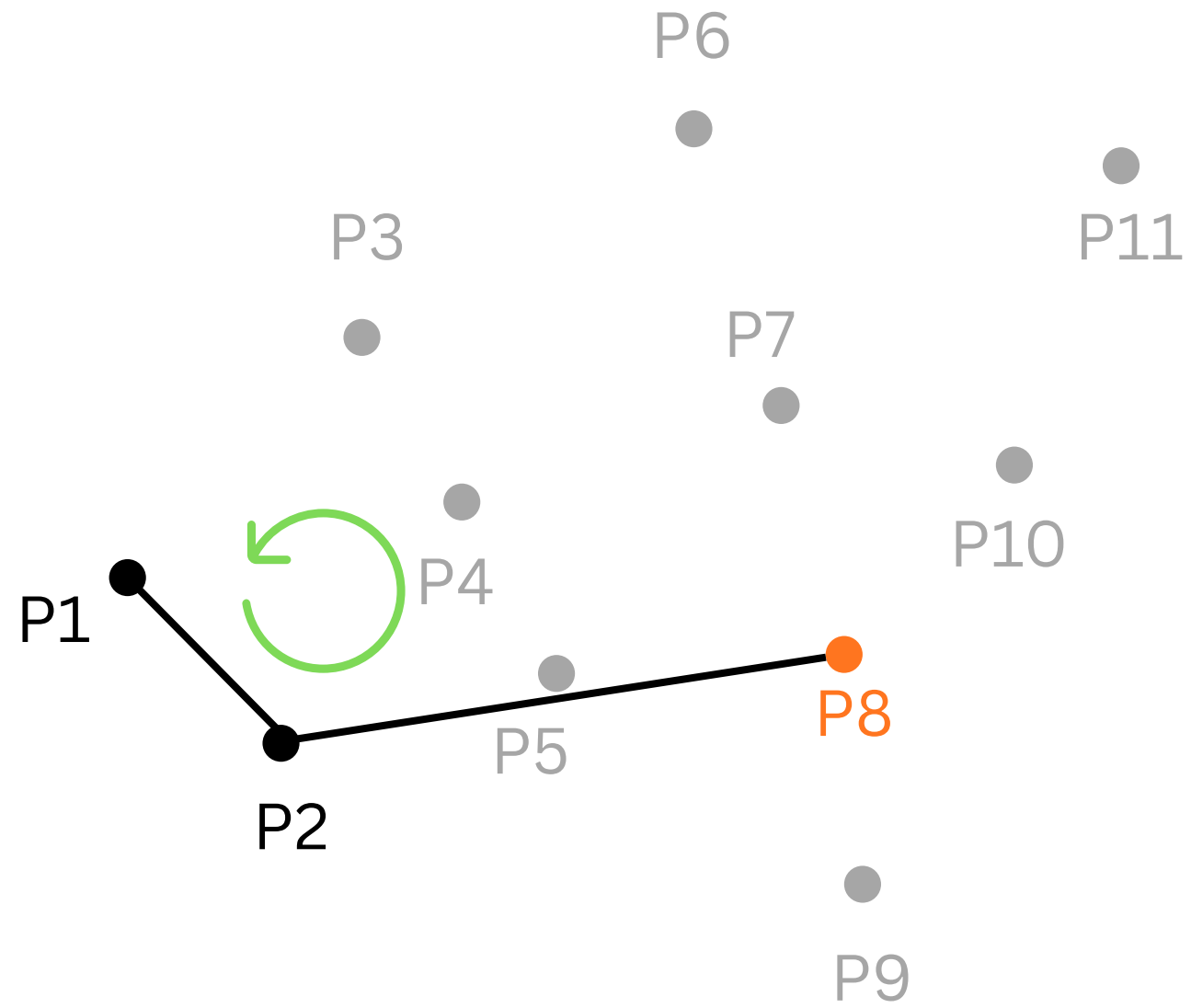
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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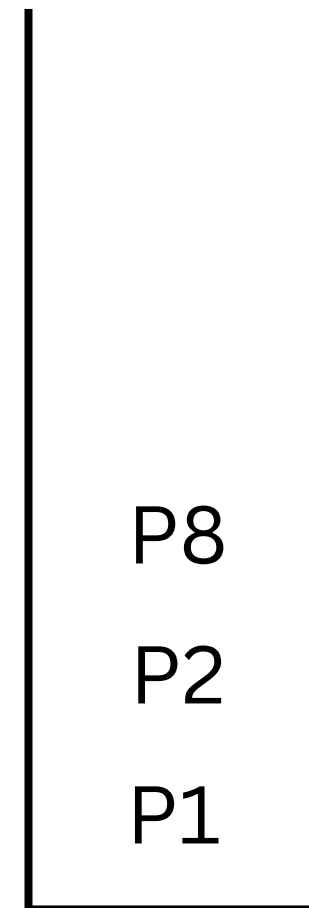
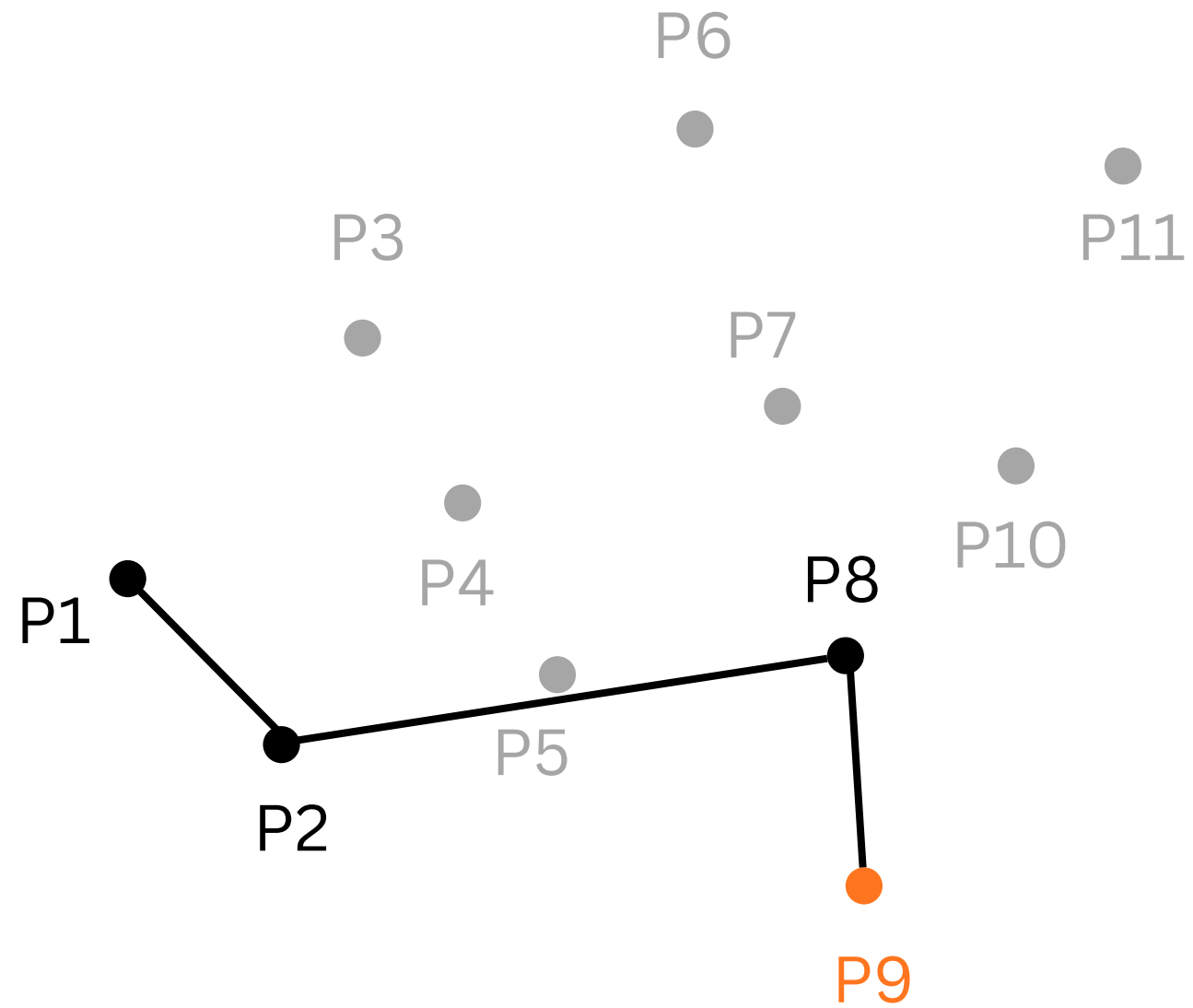
P1 P2 P3 P4 P5 P6 P7 **P8** P9 P10 P11



Stack

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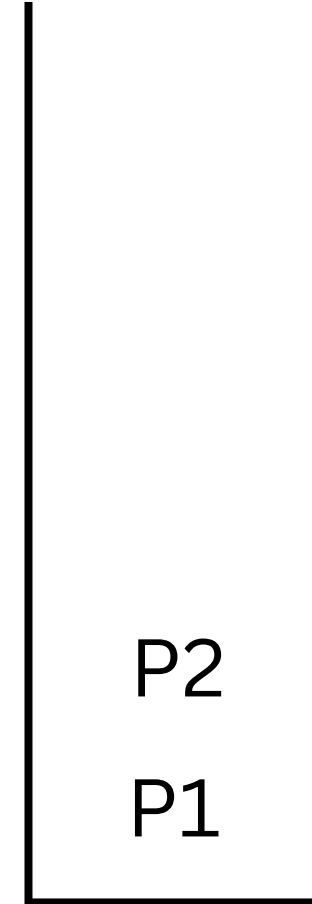
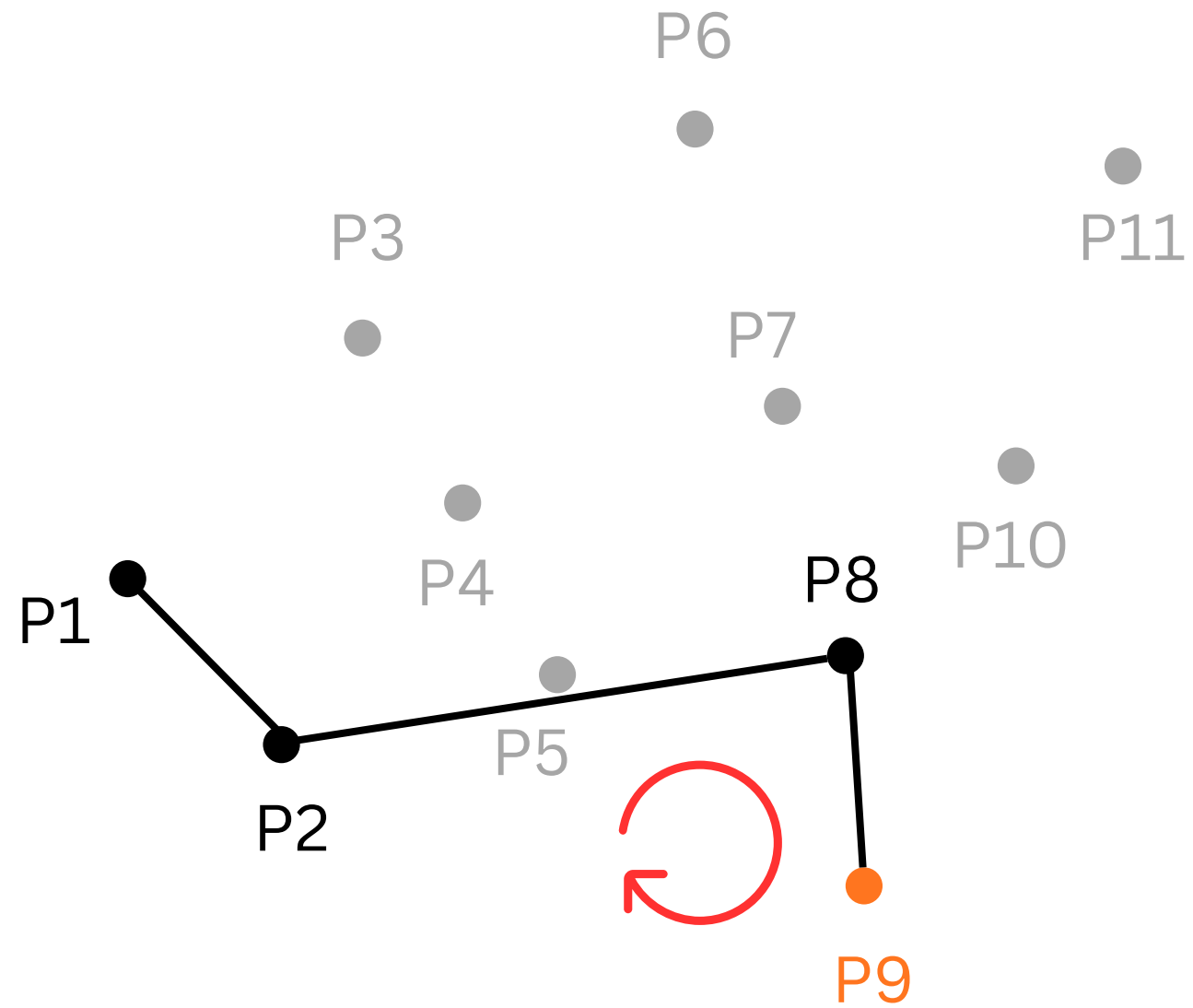
P1 P2 P3 P4 P5 P6 P7 P8 **P9** P10 P11



Stack

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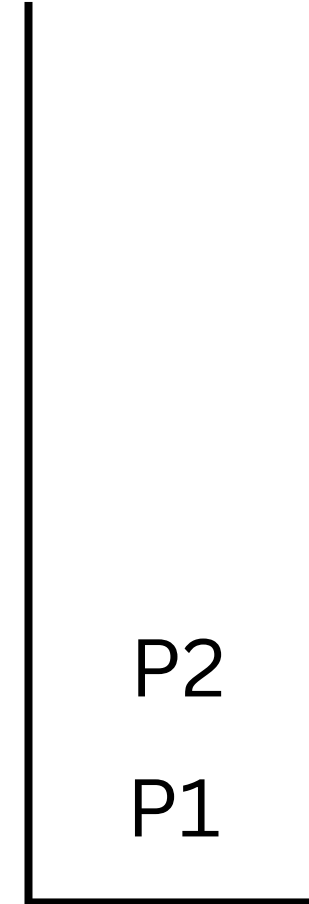
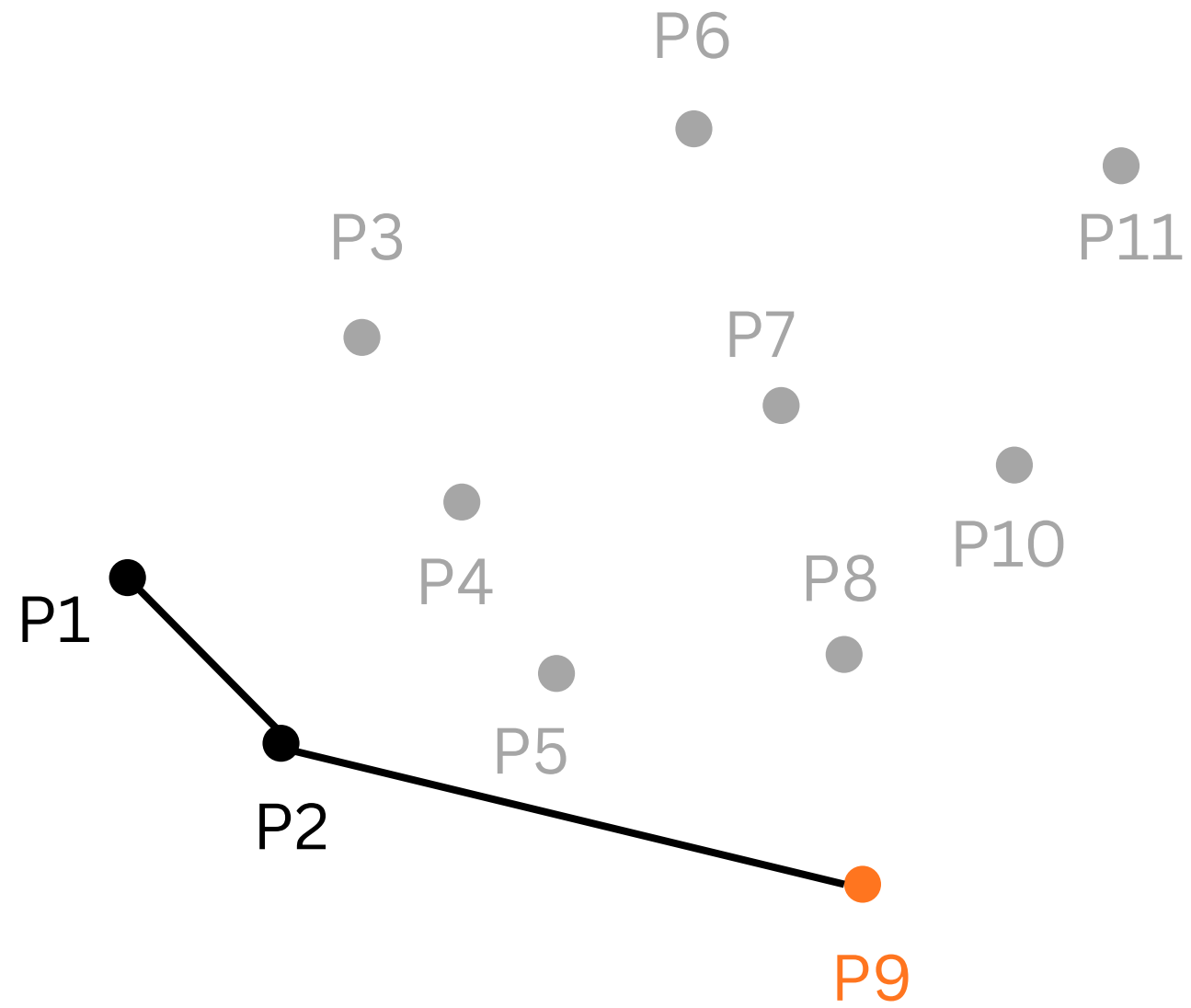
P1 P2 P3 P4 P5 P6 P7 P8 **P9** P10 P11



Stack

Array

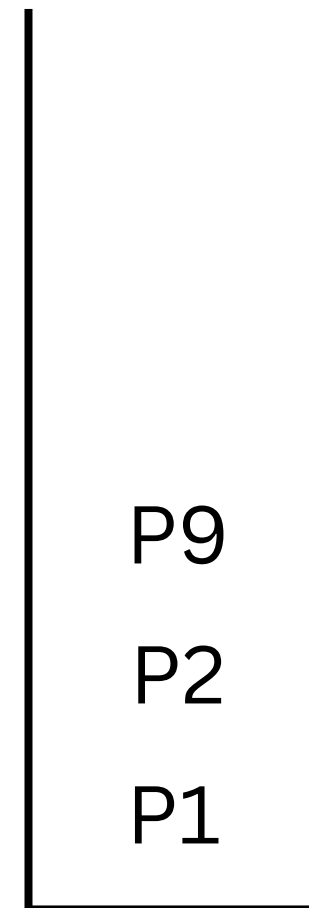
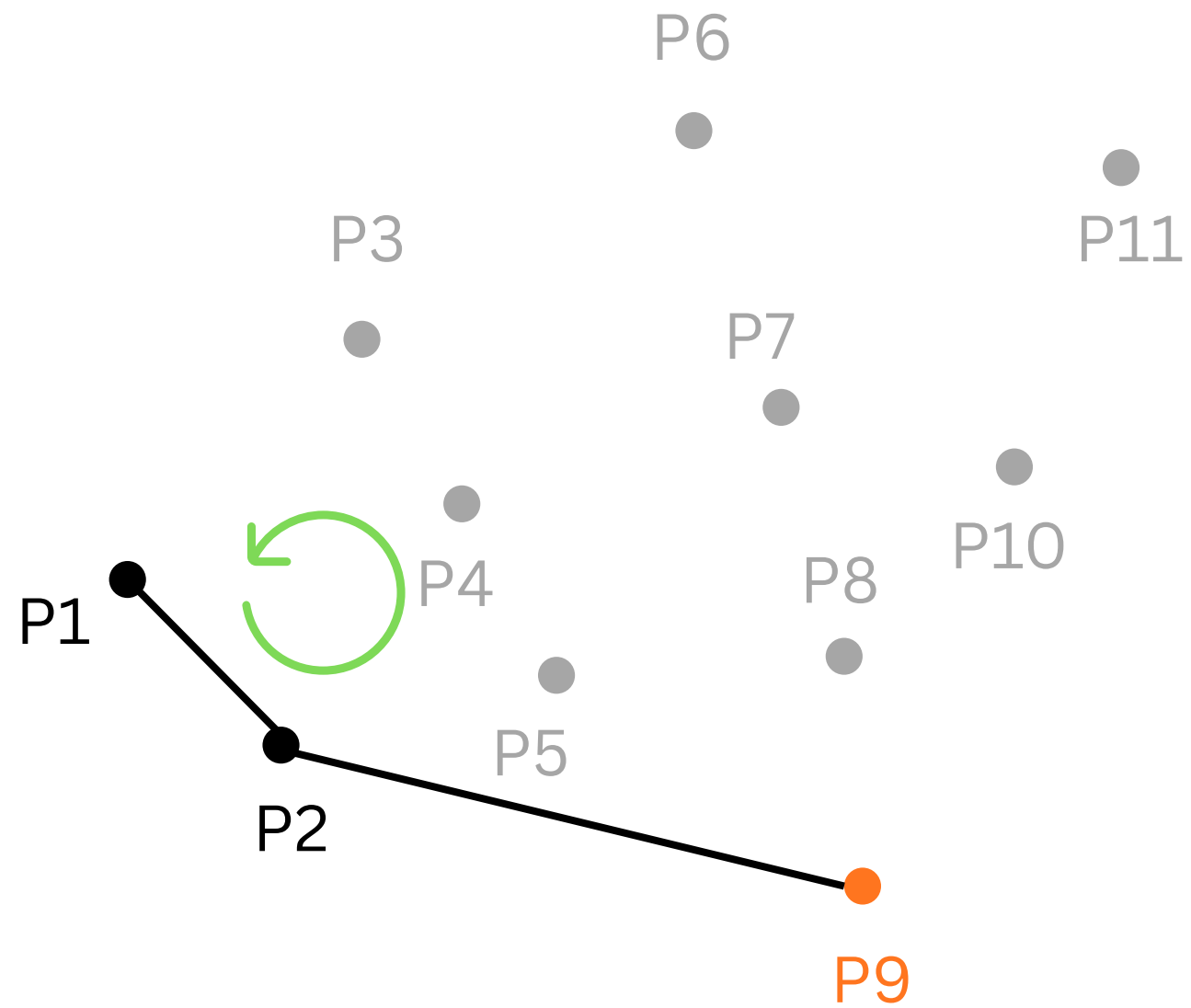
P1 P2 P3 P4 P5 P6 P7 P8 **P9** P10 P11



Stack

Array

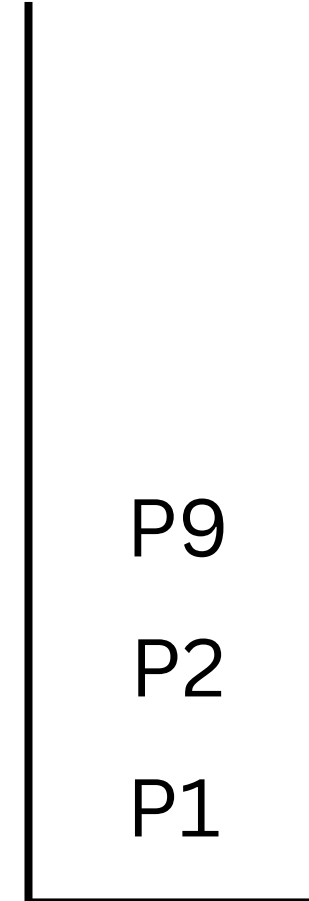
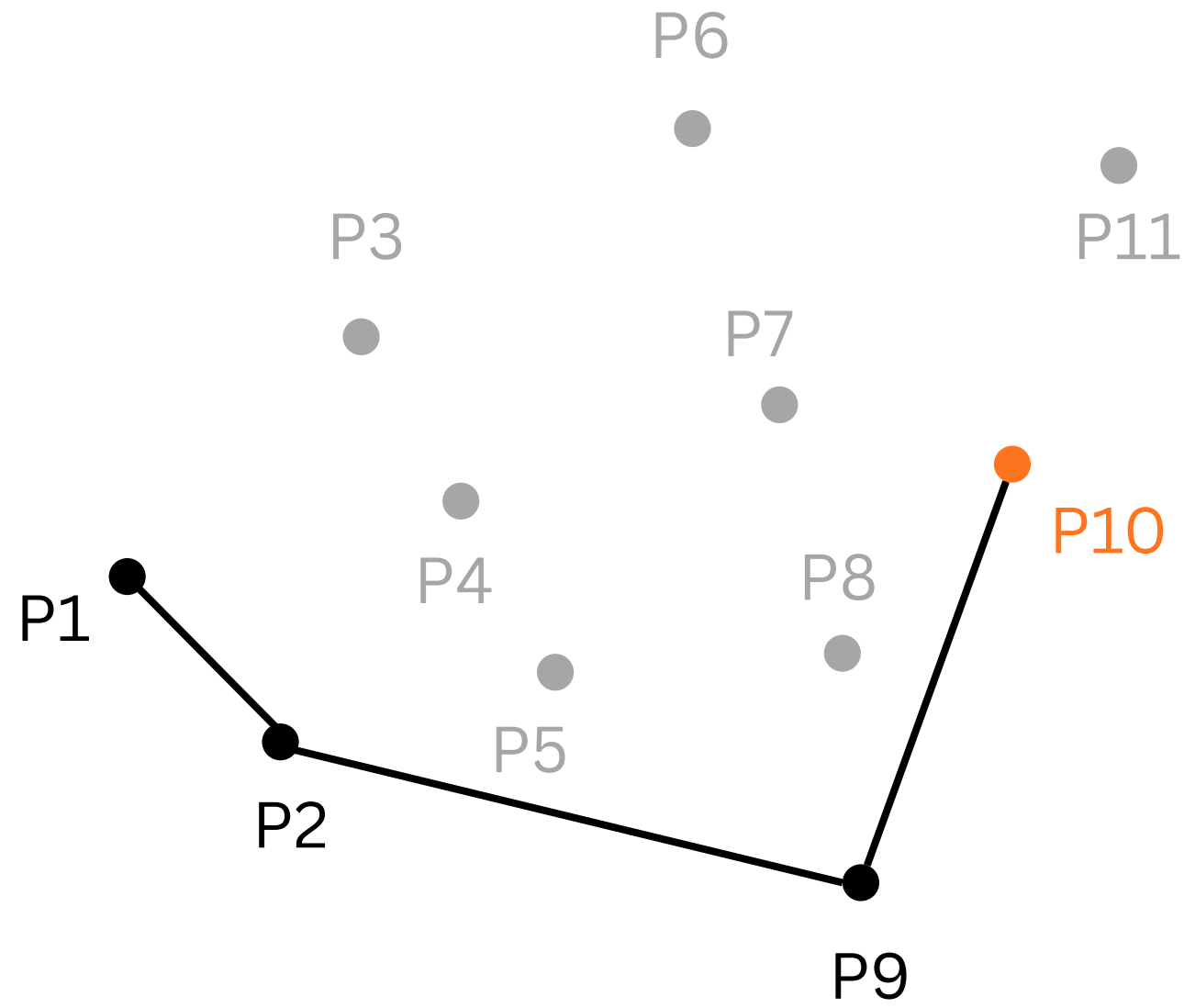
P1 P2 P3 P4 P5 P6 P7 P8 **P9** P10 P11



Stack

Array

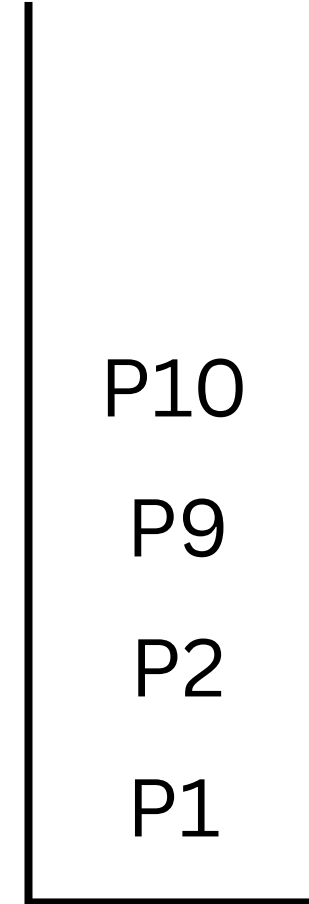
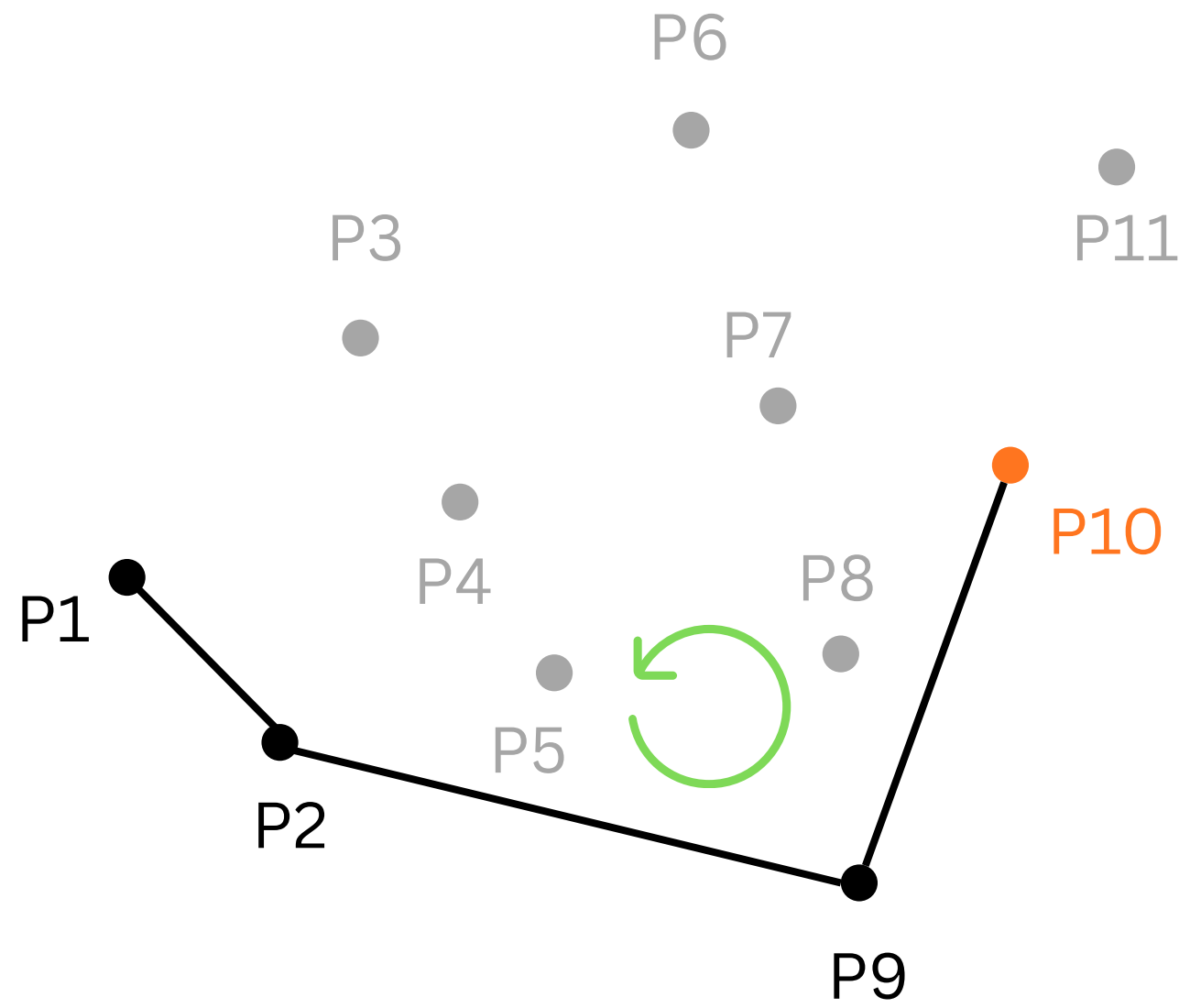
P1 P2 P3 P4 P5 P6 P7 P8 P9 **P10** P11



Stack

Array

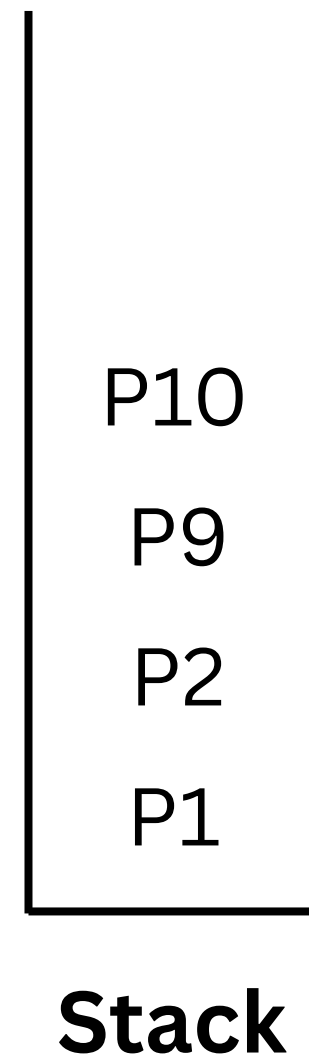
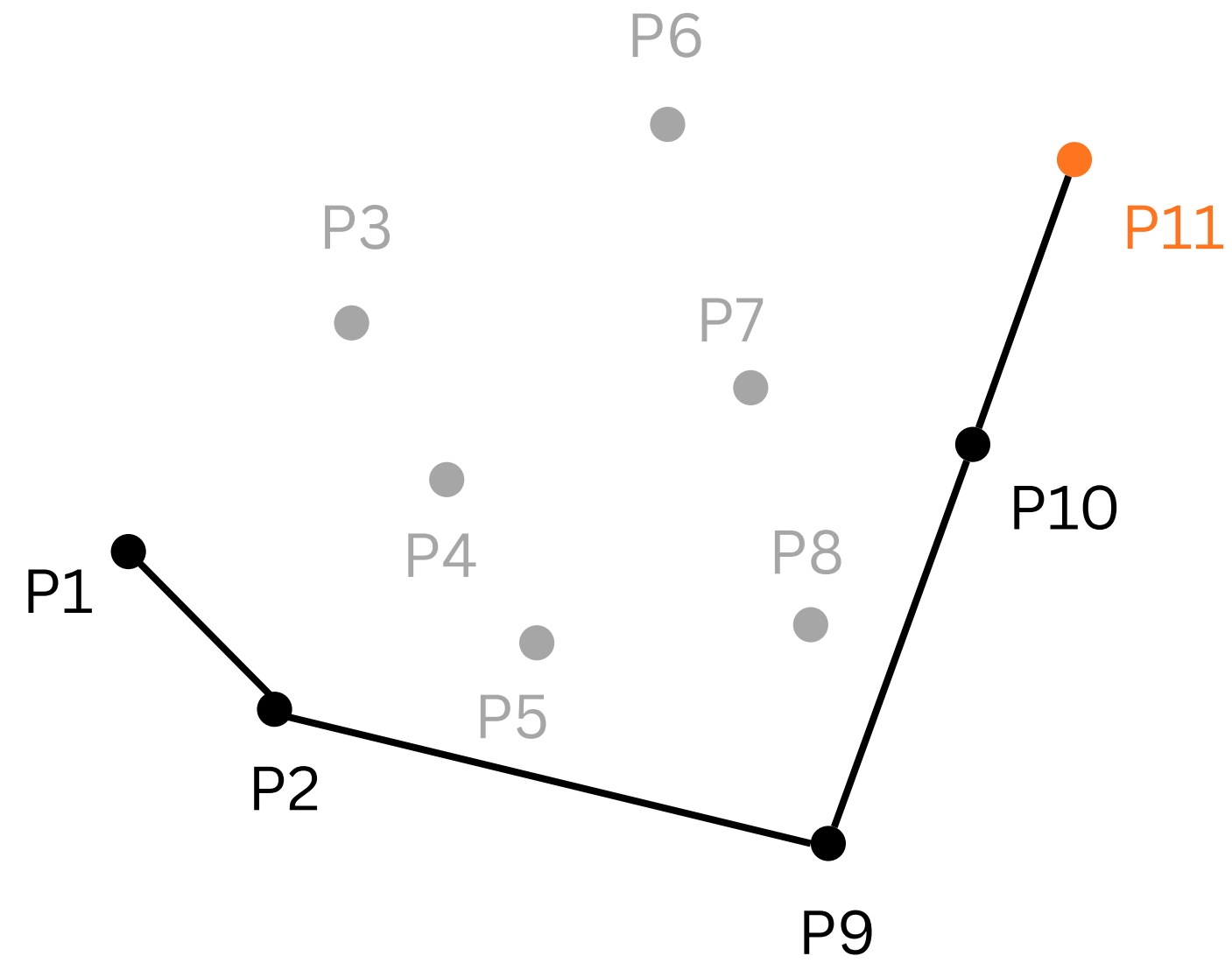
P1 P2 P3 P4 P5 P6 P7 P8 P9 **P10** P11



Stack

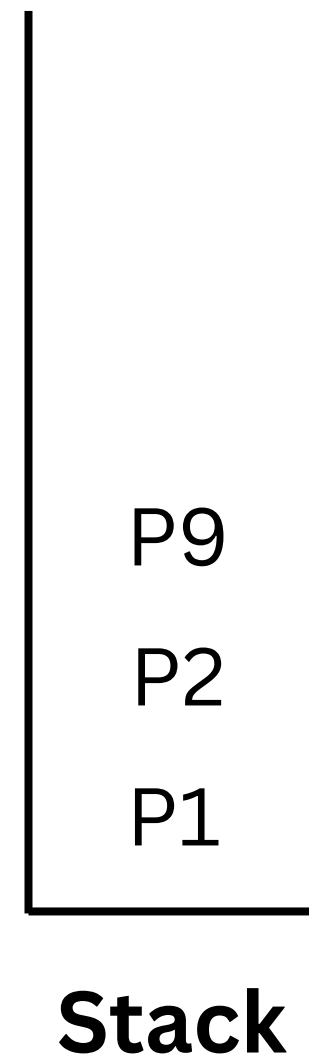
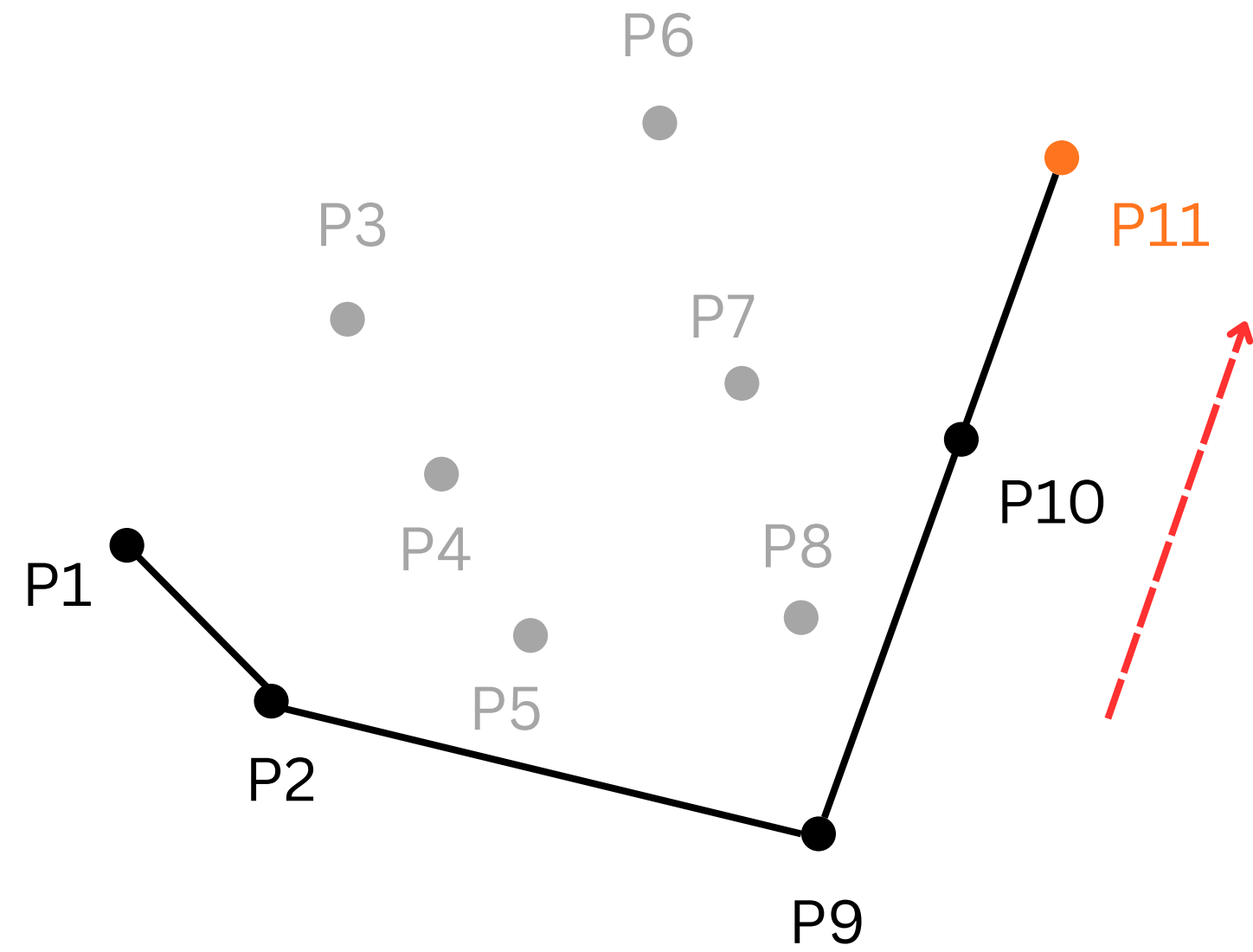
Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 **P11**
↑



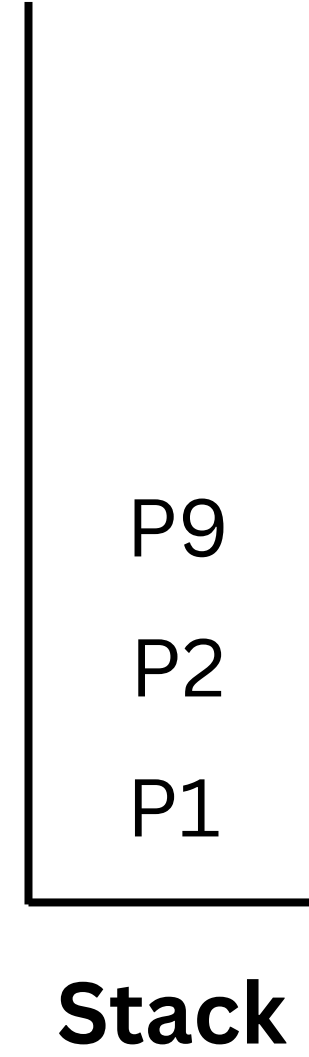
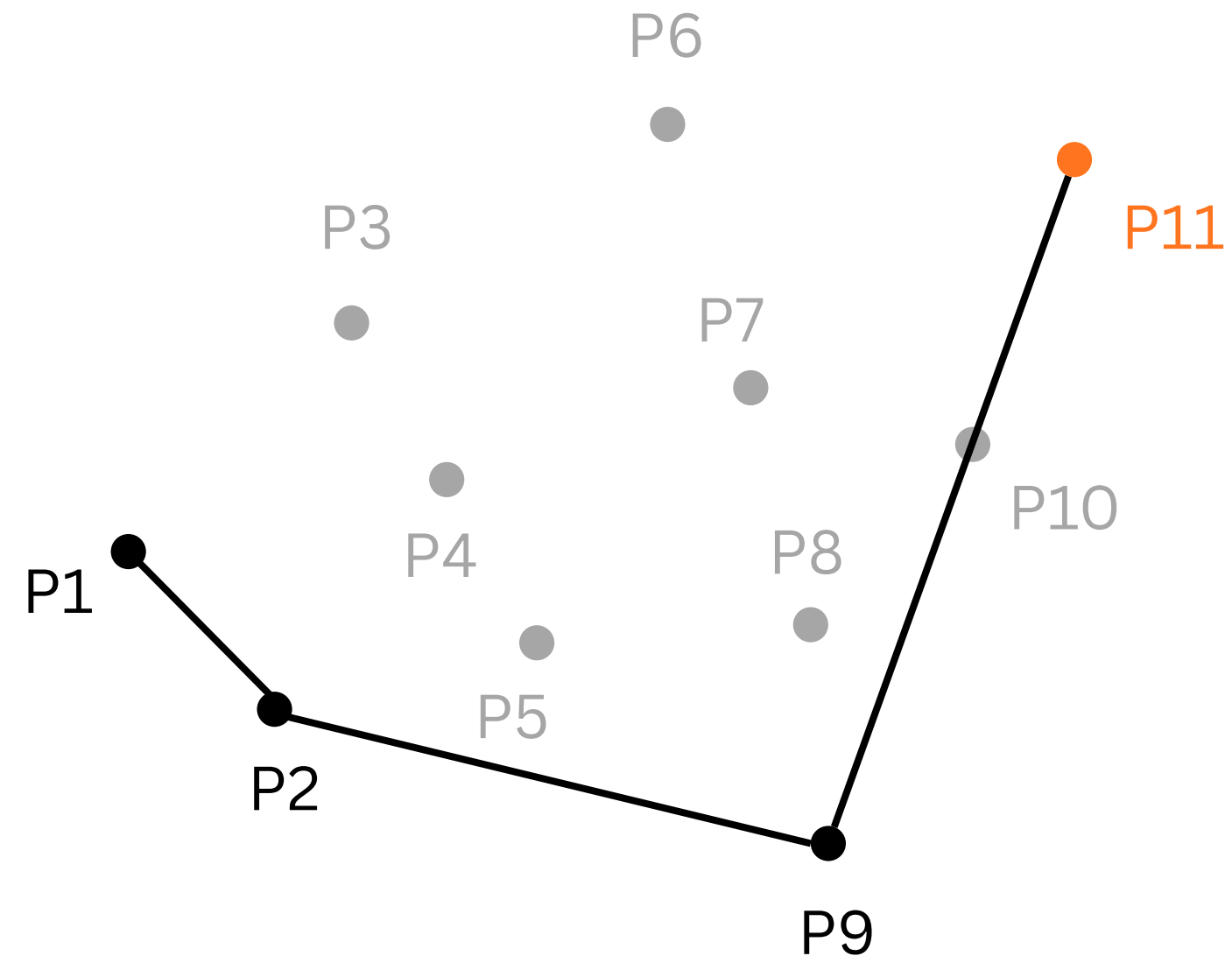
Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 **P11**
↑



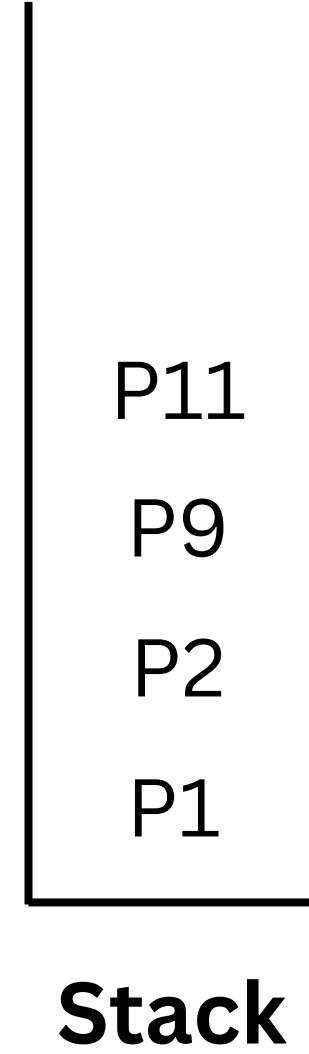
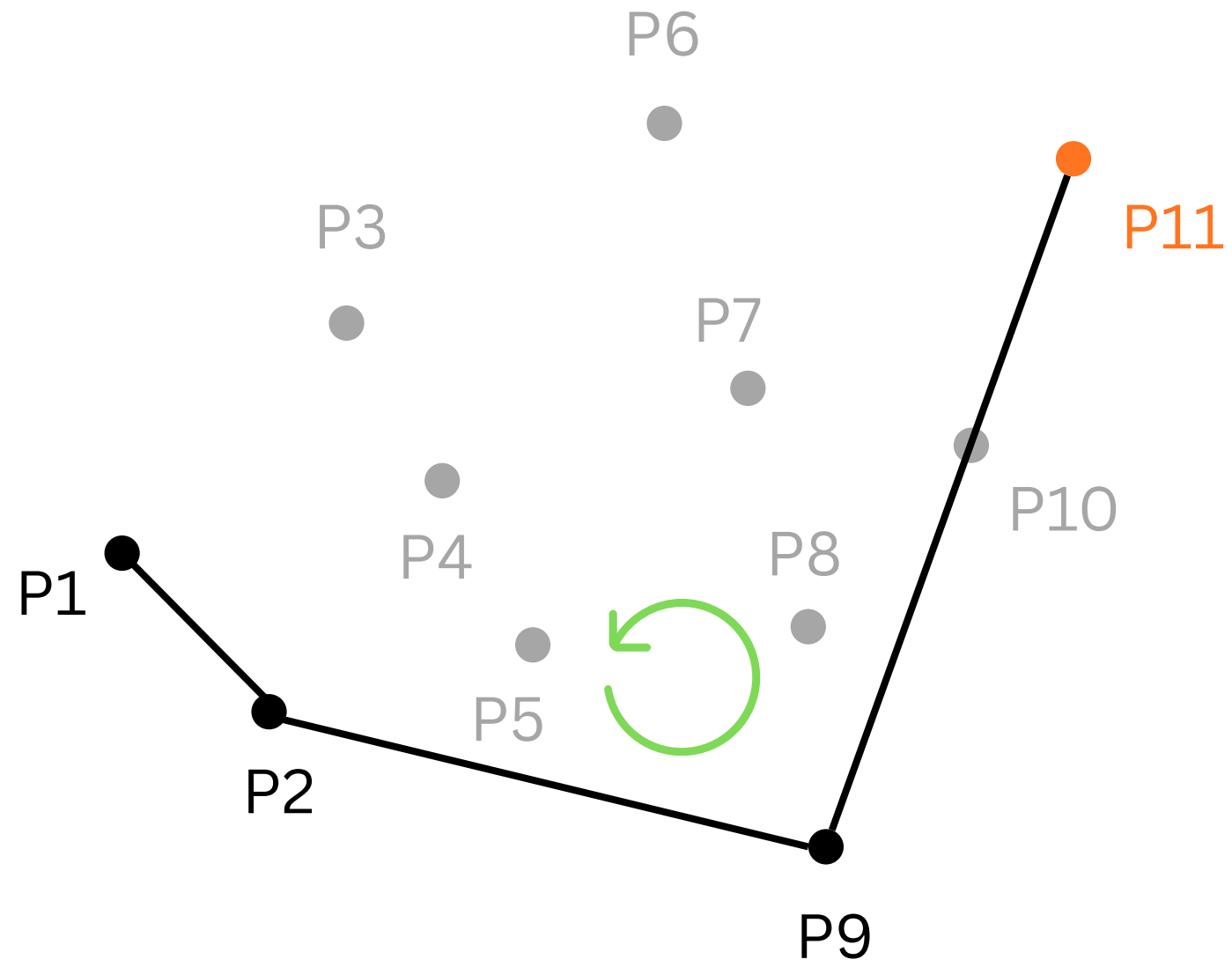
Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 **P11**
↑



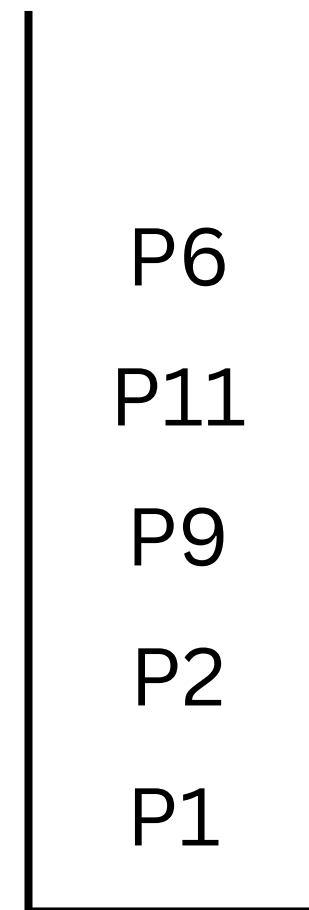
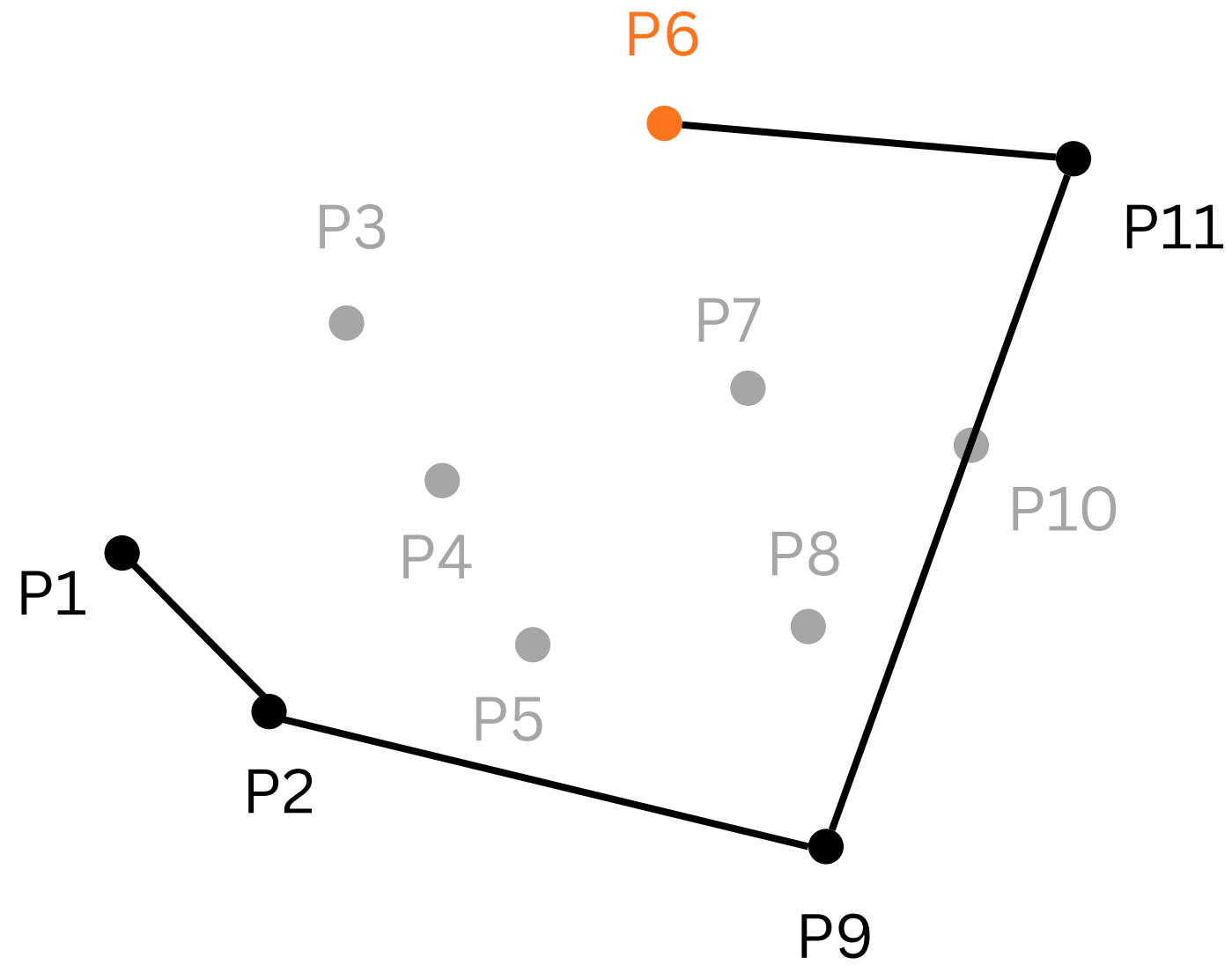
Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 **P11**
↑



Array

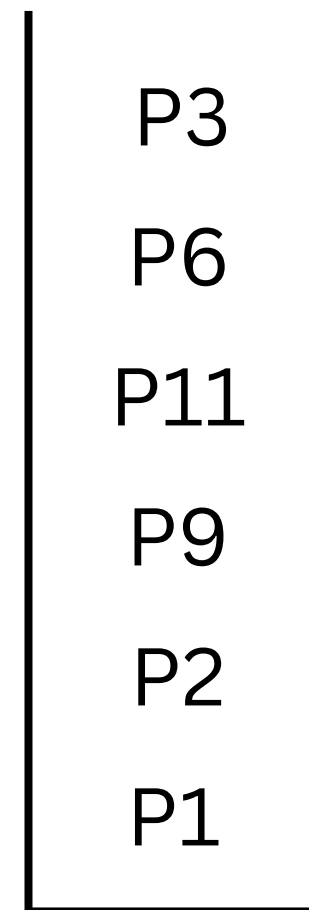
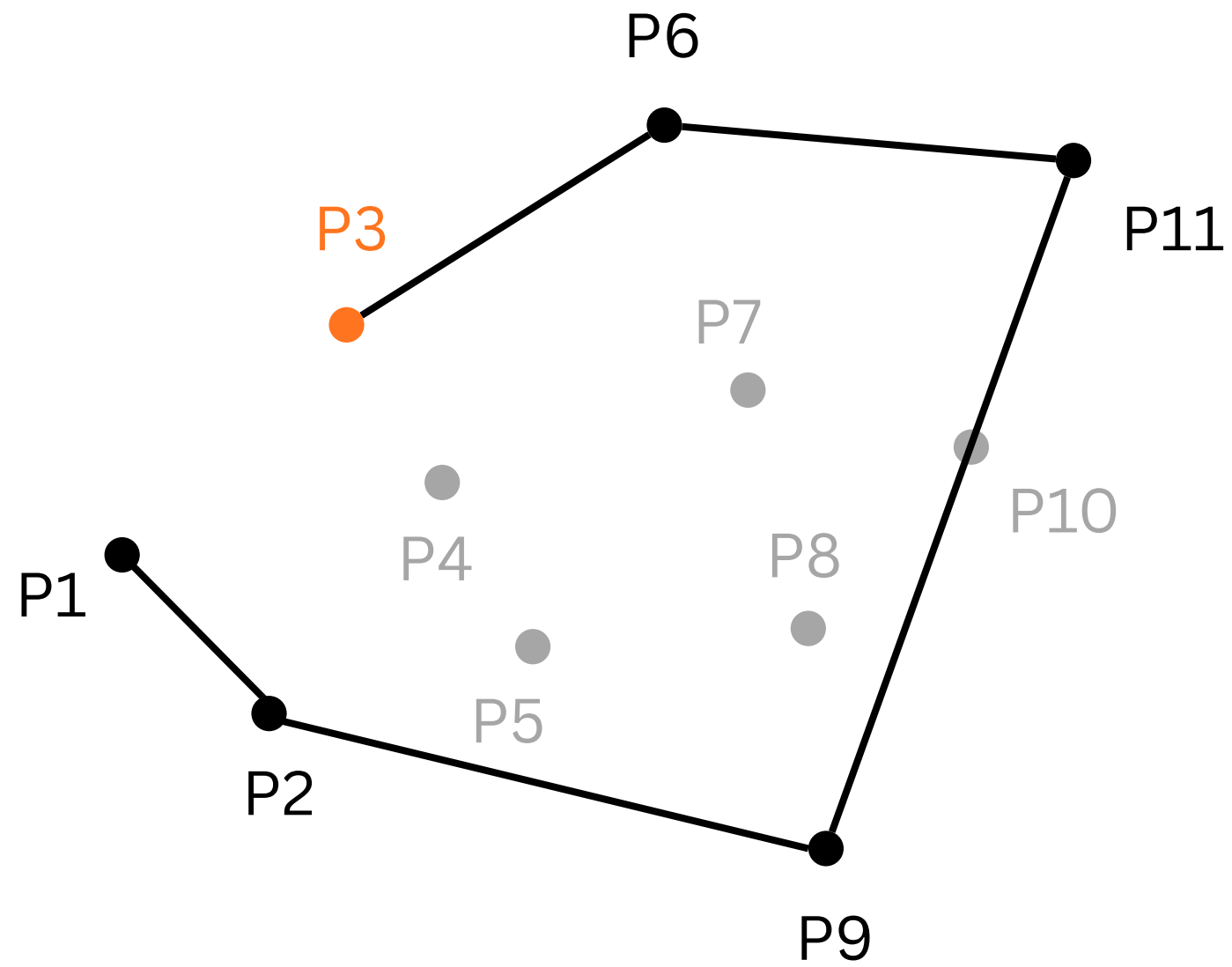
P1 P2 P3 P4 P5 **P6** P7 P8 P9 P10 P11



Stack

Array

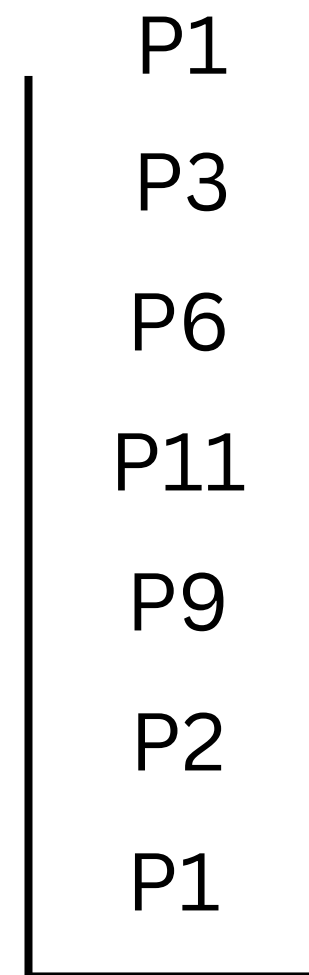
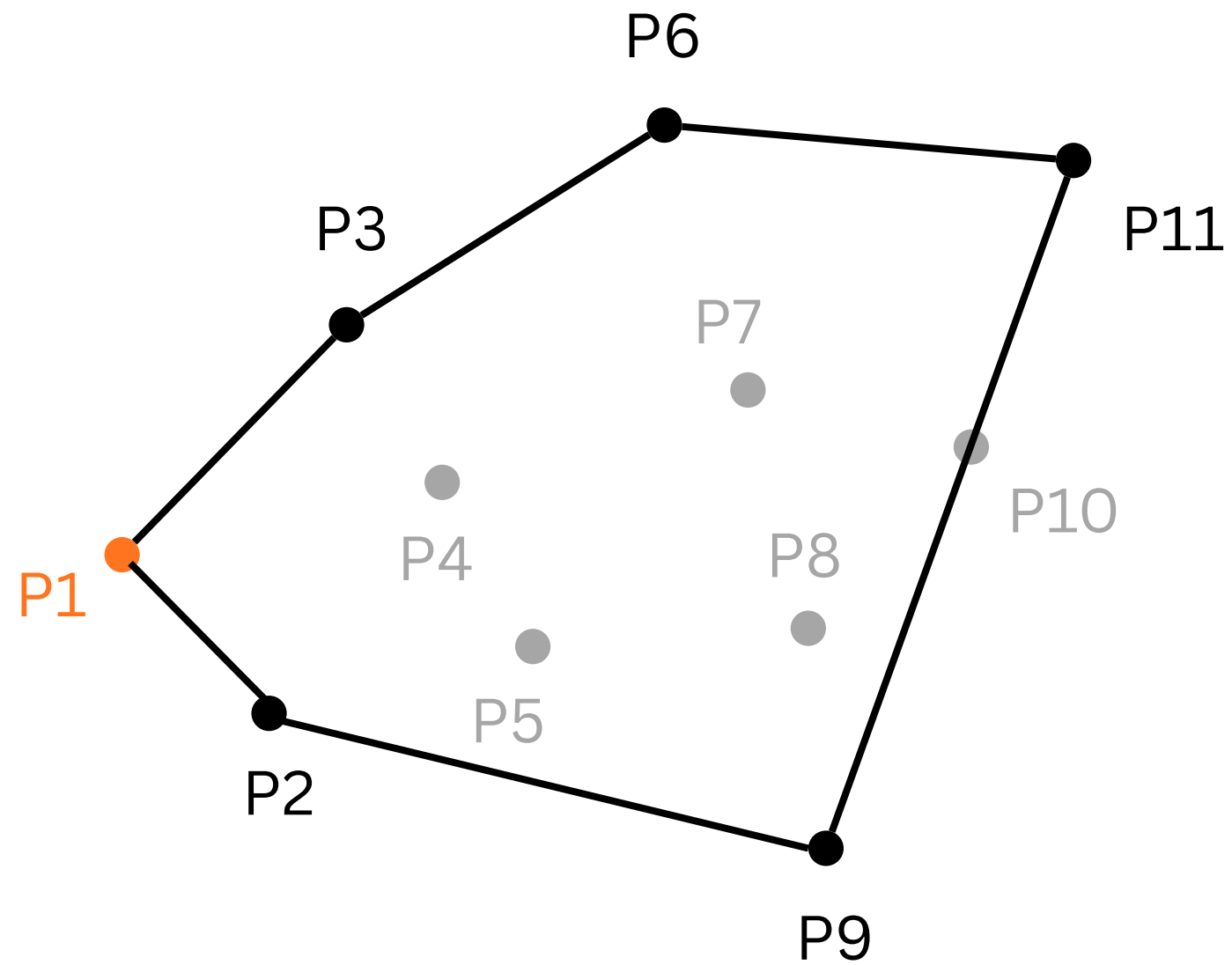
P1 P2 **P3** P4 P5 P6 P7 P8 P9 P10 P11



Stack

Array

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 P11



Stack

